

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-12 23:45:29

Data & configuration

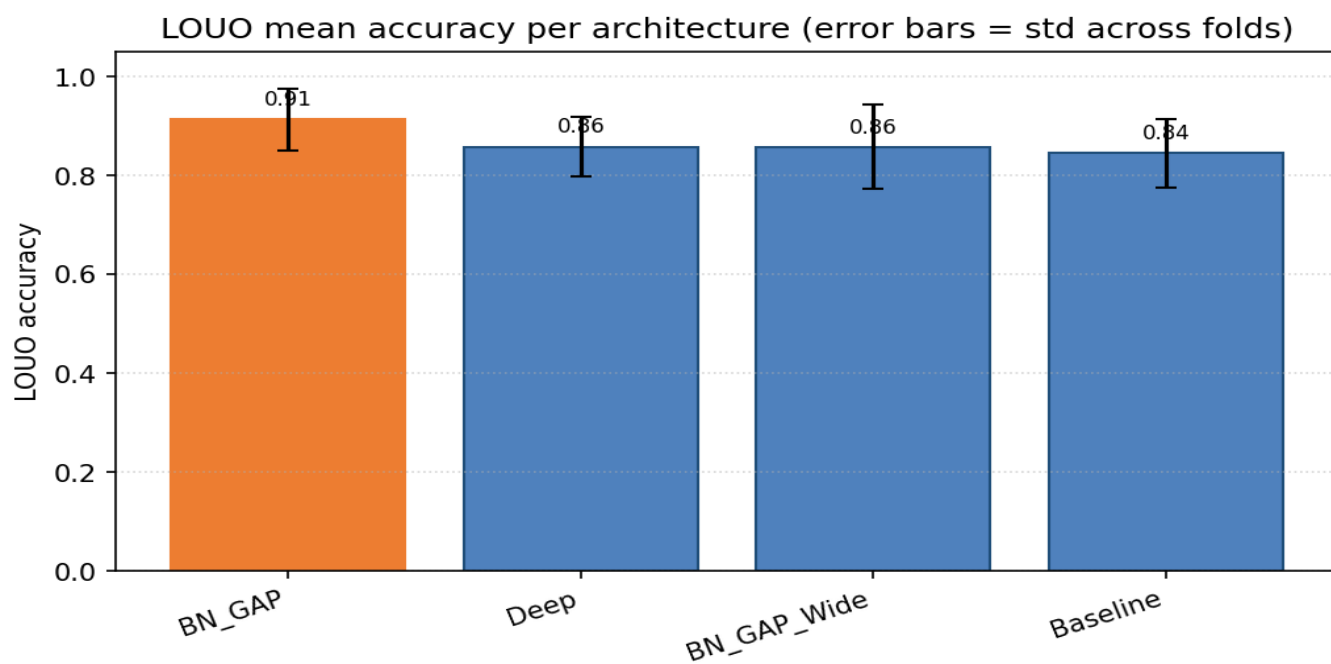
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	11 (11_Mansh, 12_Marcus, 14_StephenV2, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	Double_Nothing
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	18 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	zscore
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	on · copies=2 · amplitude_scale, baseline_offset
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	BN_GAP_Wide, Deep, Baseline, BN_GAP
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

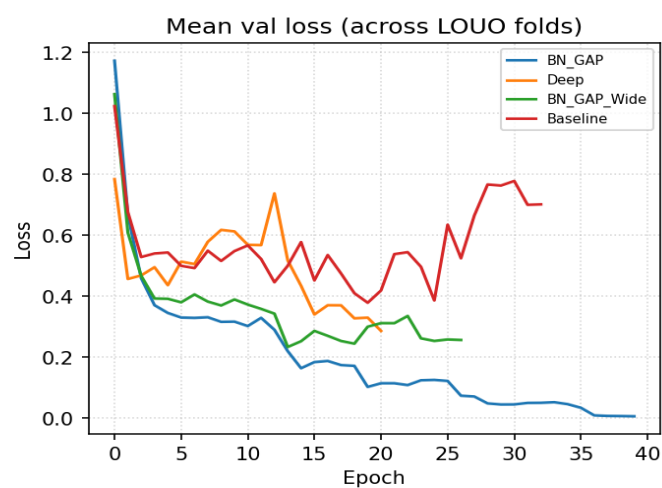
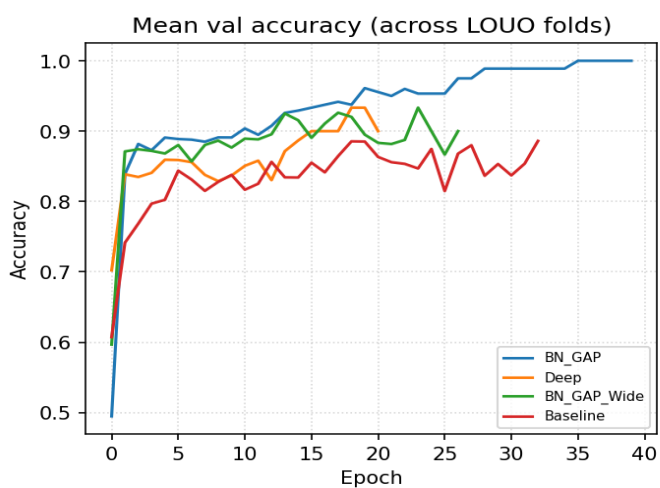
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
BN_GAP	36,966	0.9127	0.0630	11	—
Deep	183,654	0.8575	0.0602	11	—
BN_GAP_Wide	63,846	0.8572	0.0851	11	—
Baseline	92,966	0.8441	0.0698	11	—

Best LOUO mean: BN_GAP

LOUO accuracy per variant

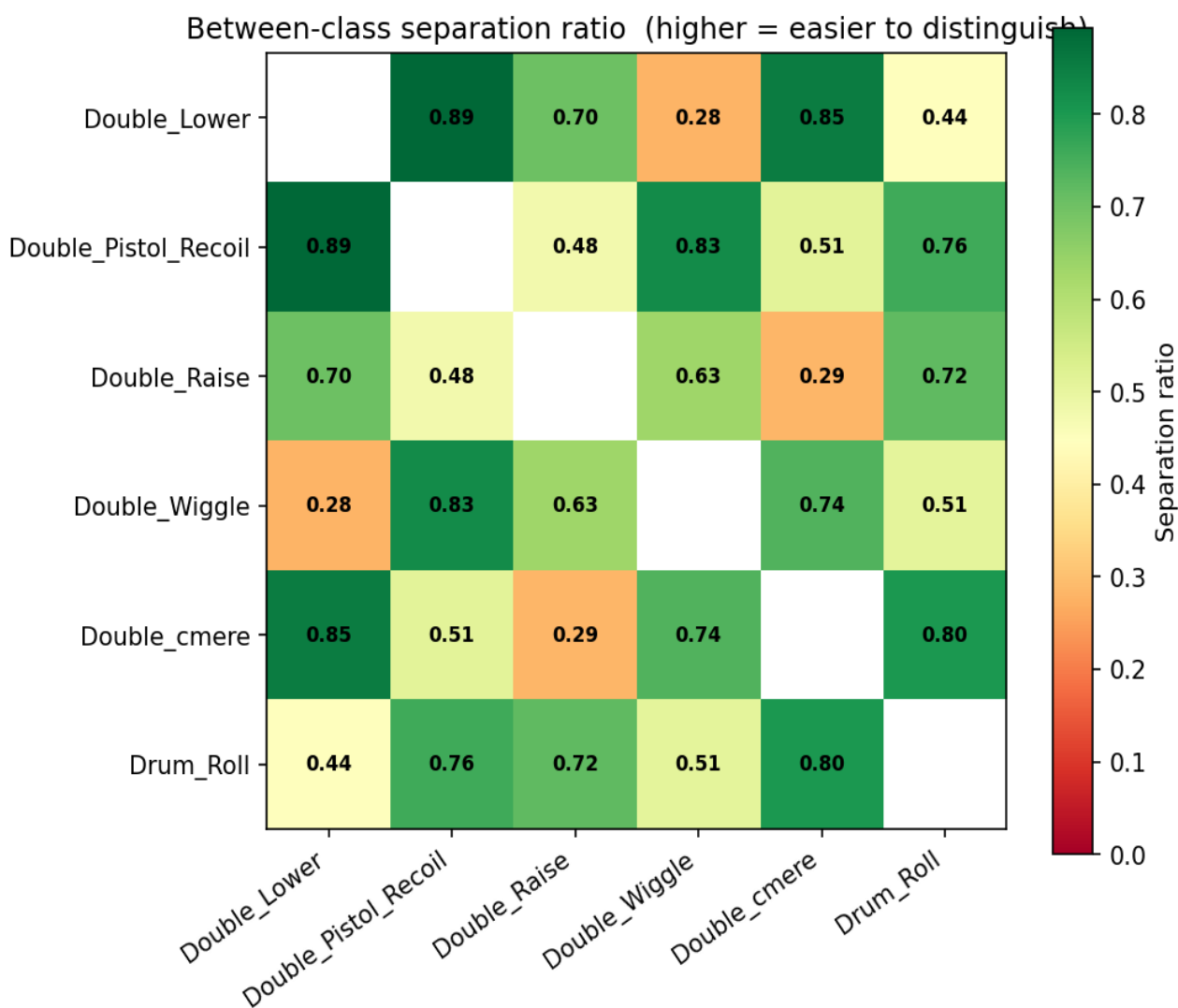
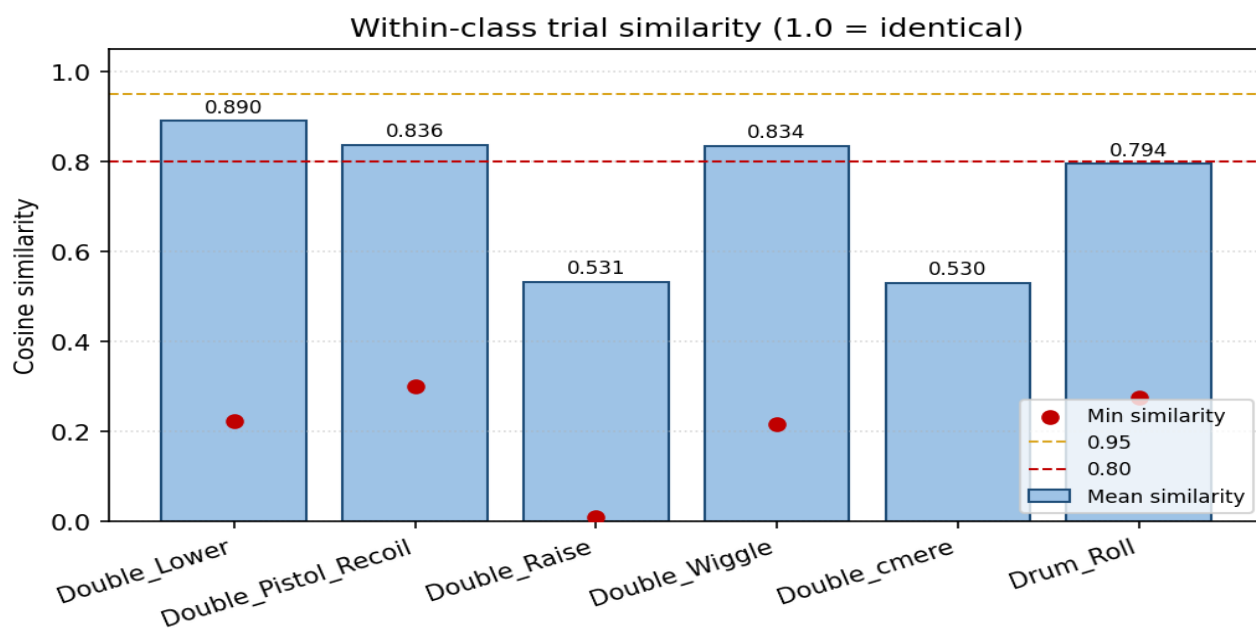


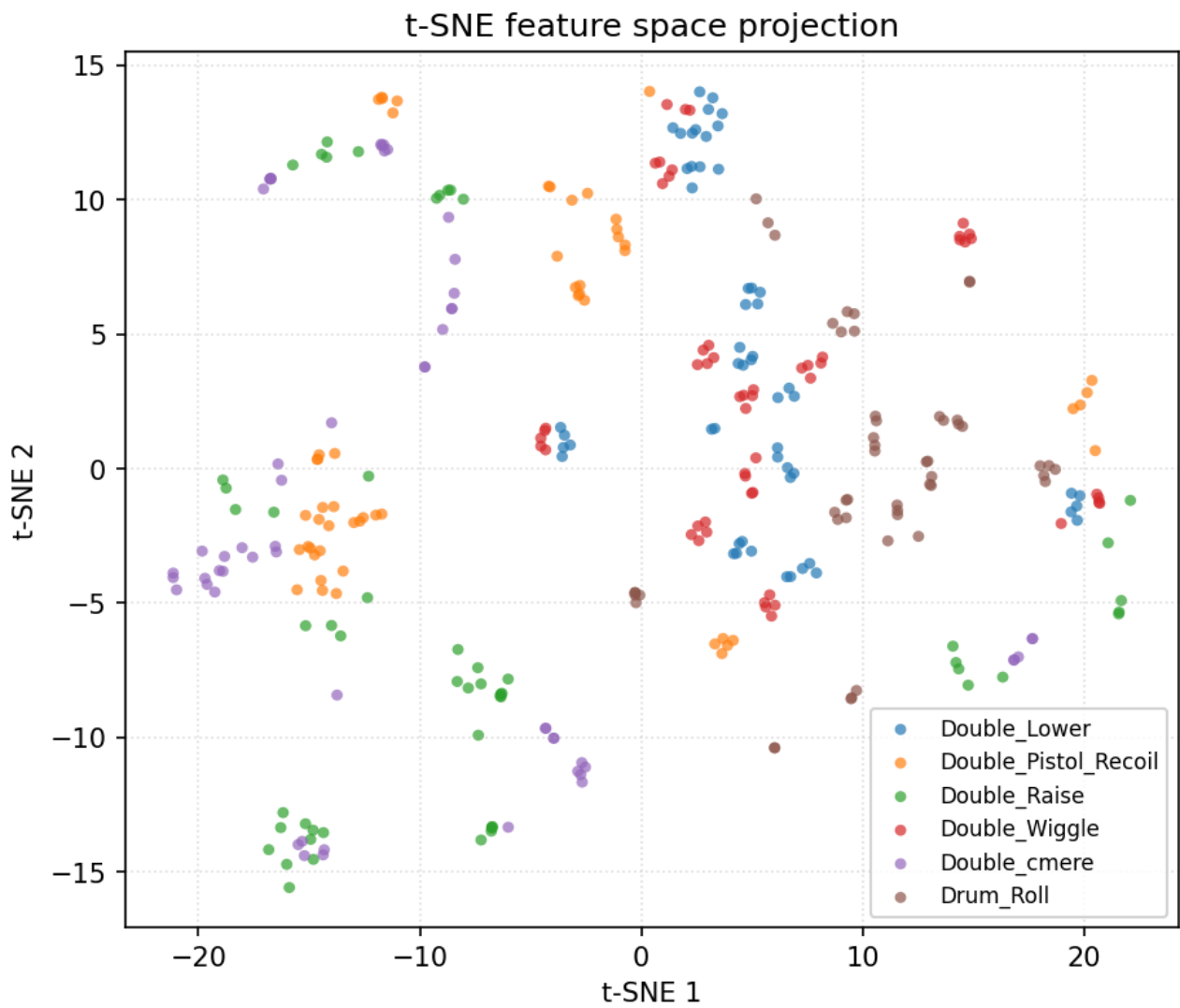
Mean training curves across LOUO folds



Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).

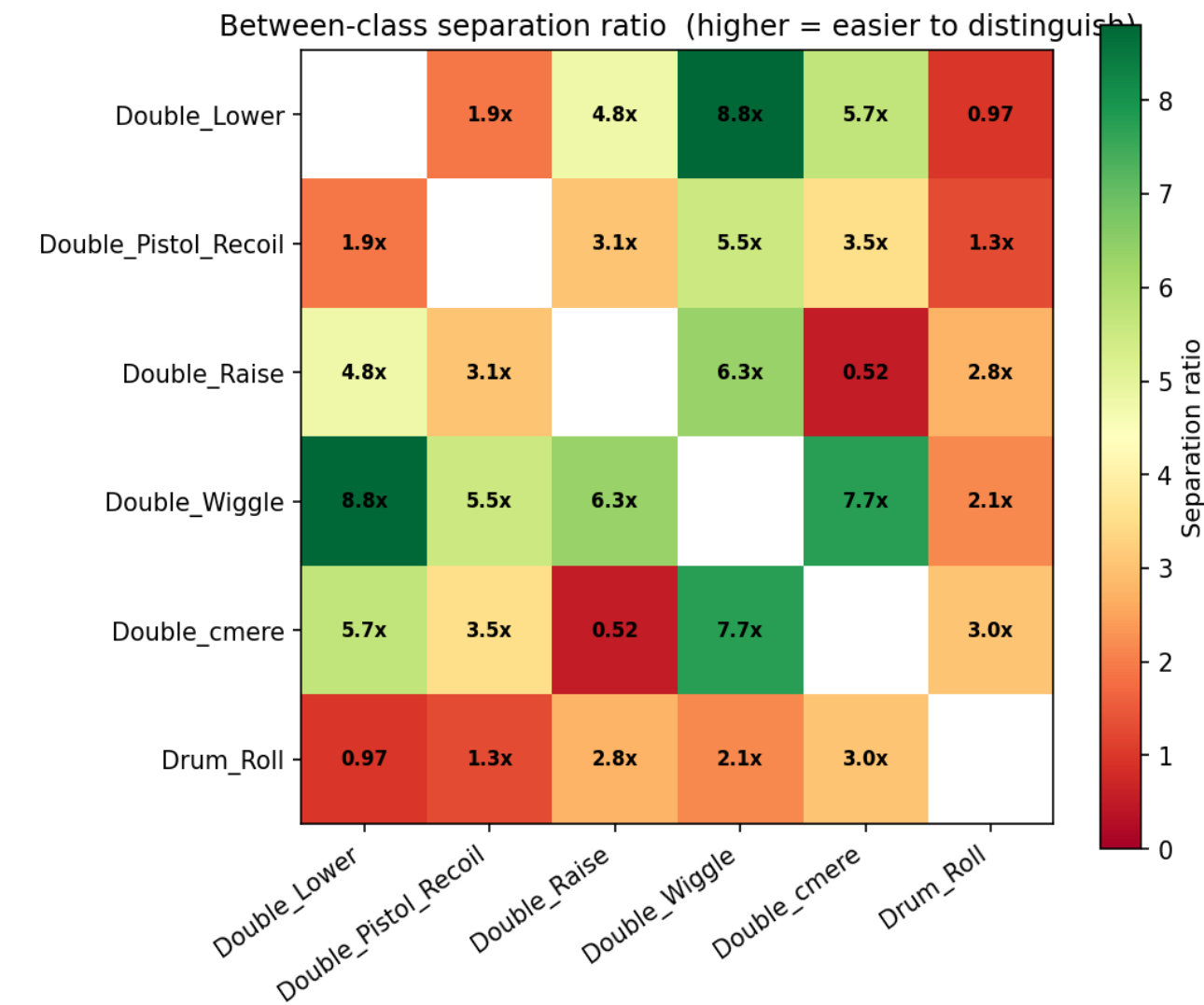
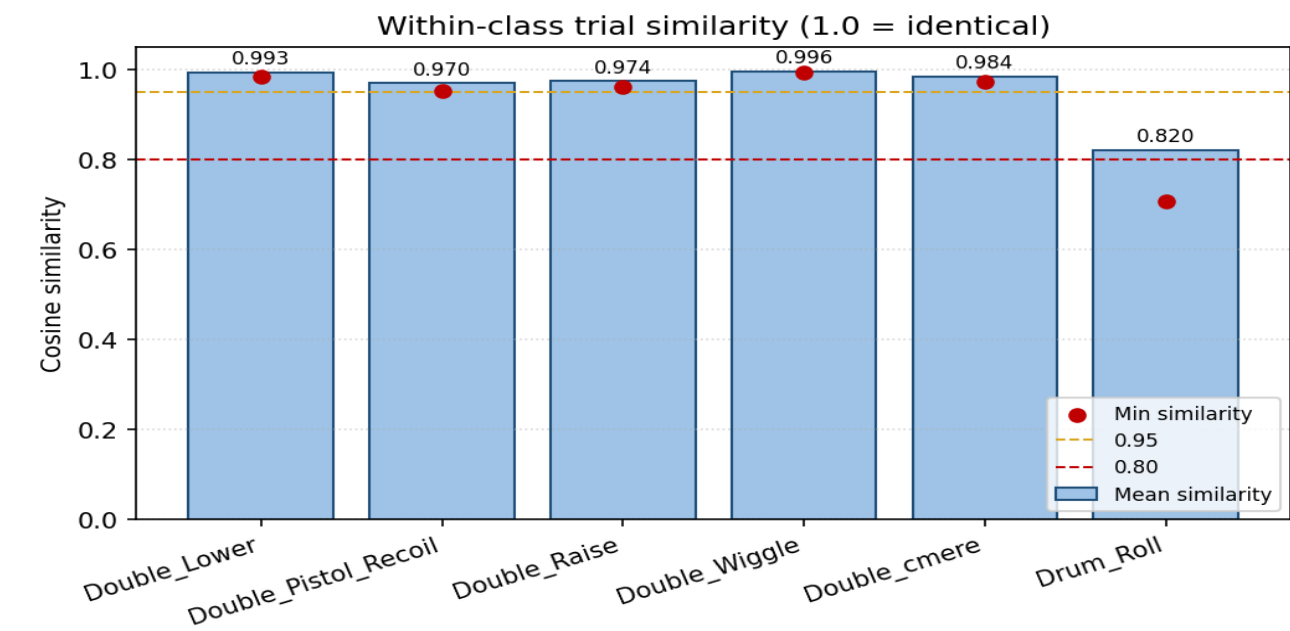




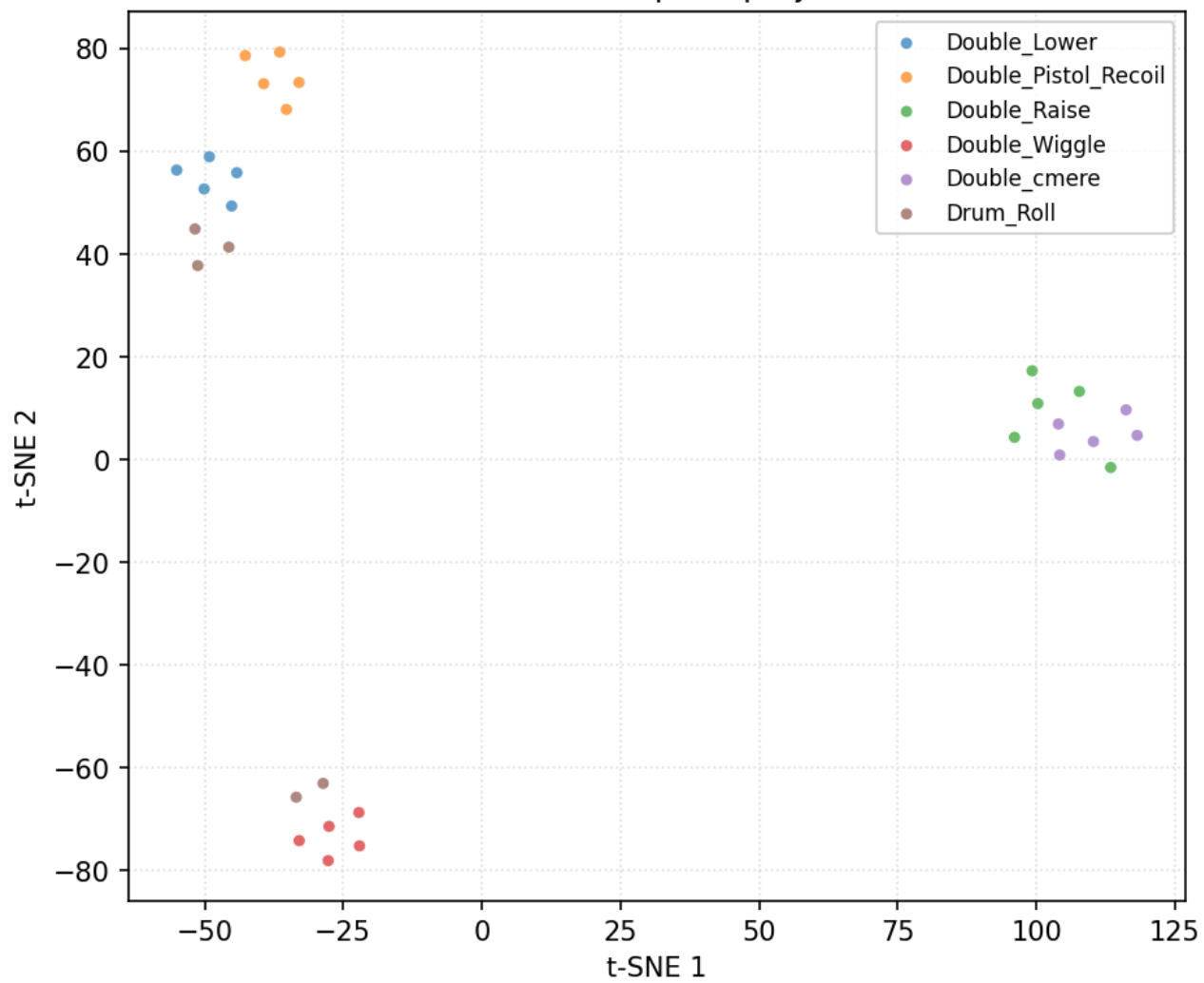
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

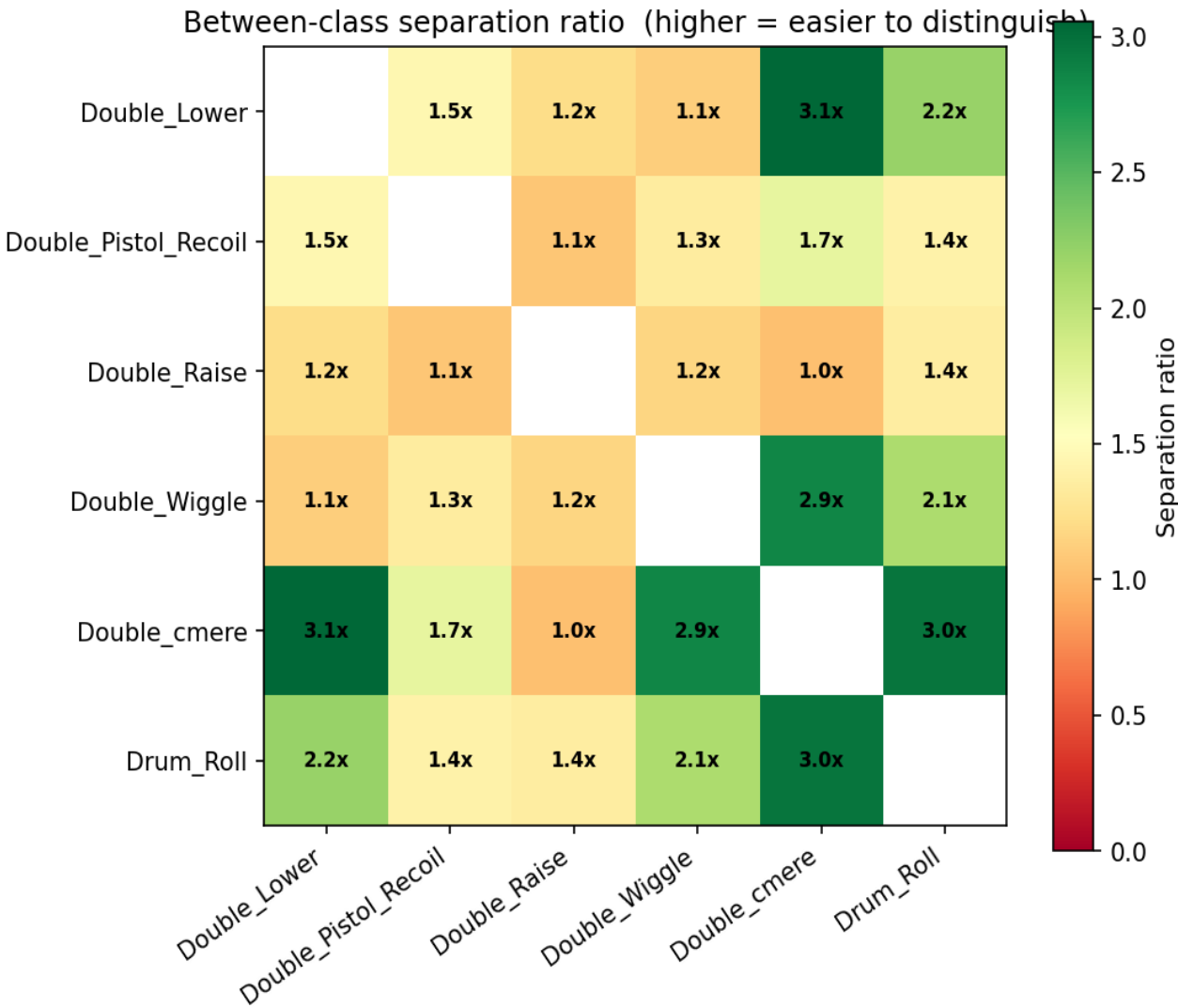
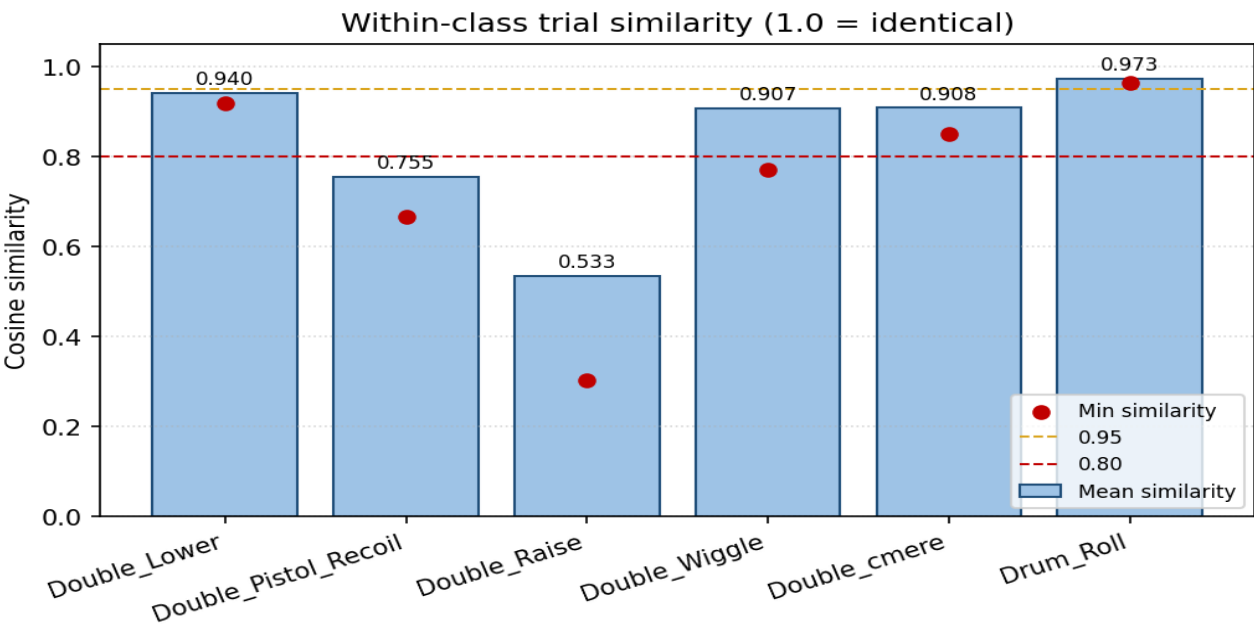


t-SNE feature space projection

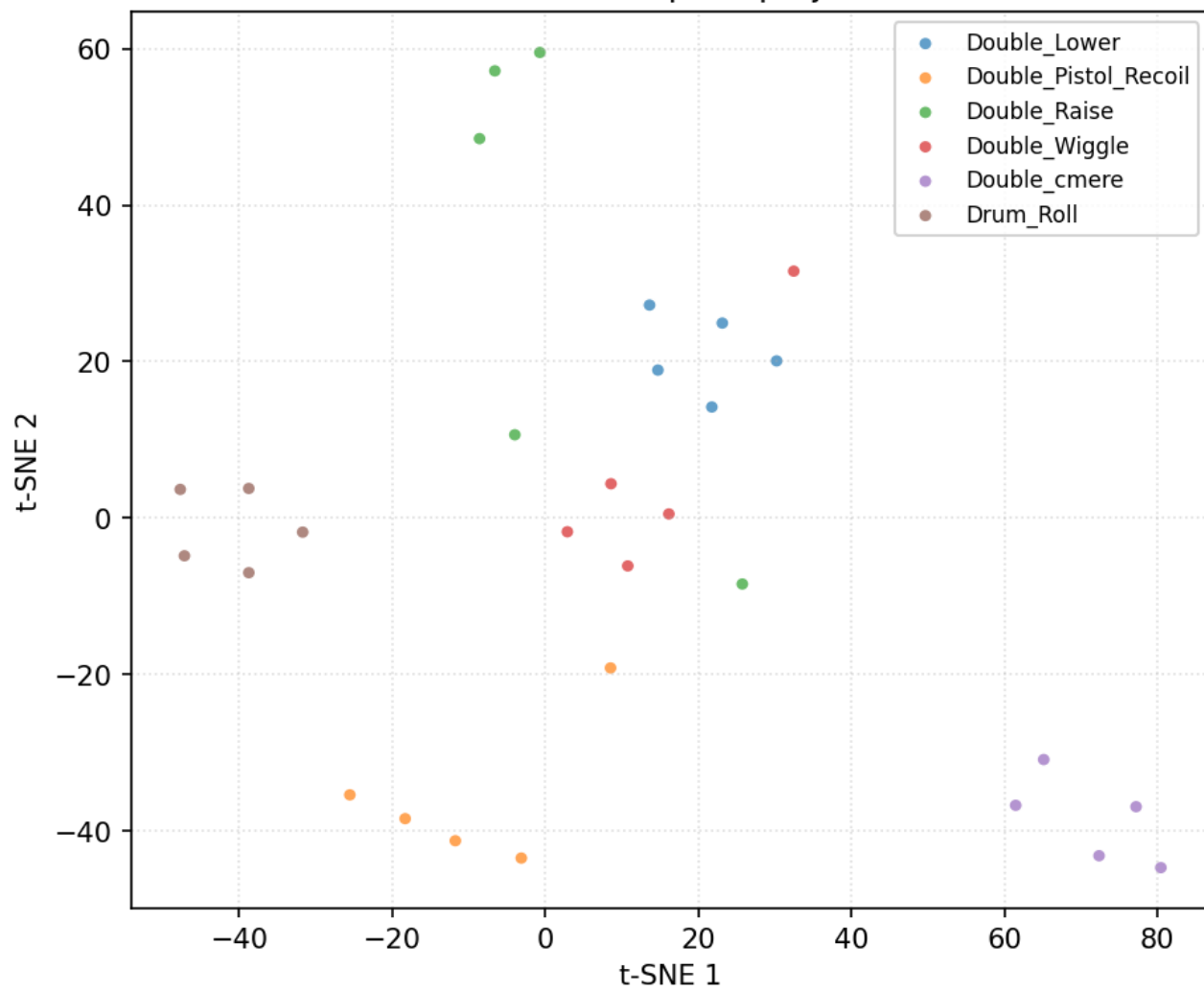


Data diagnostics — user 12_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

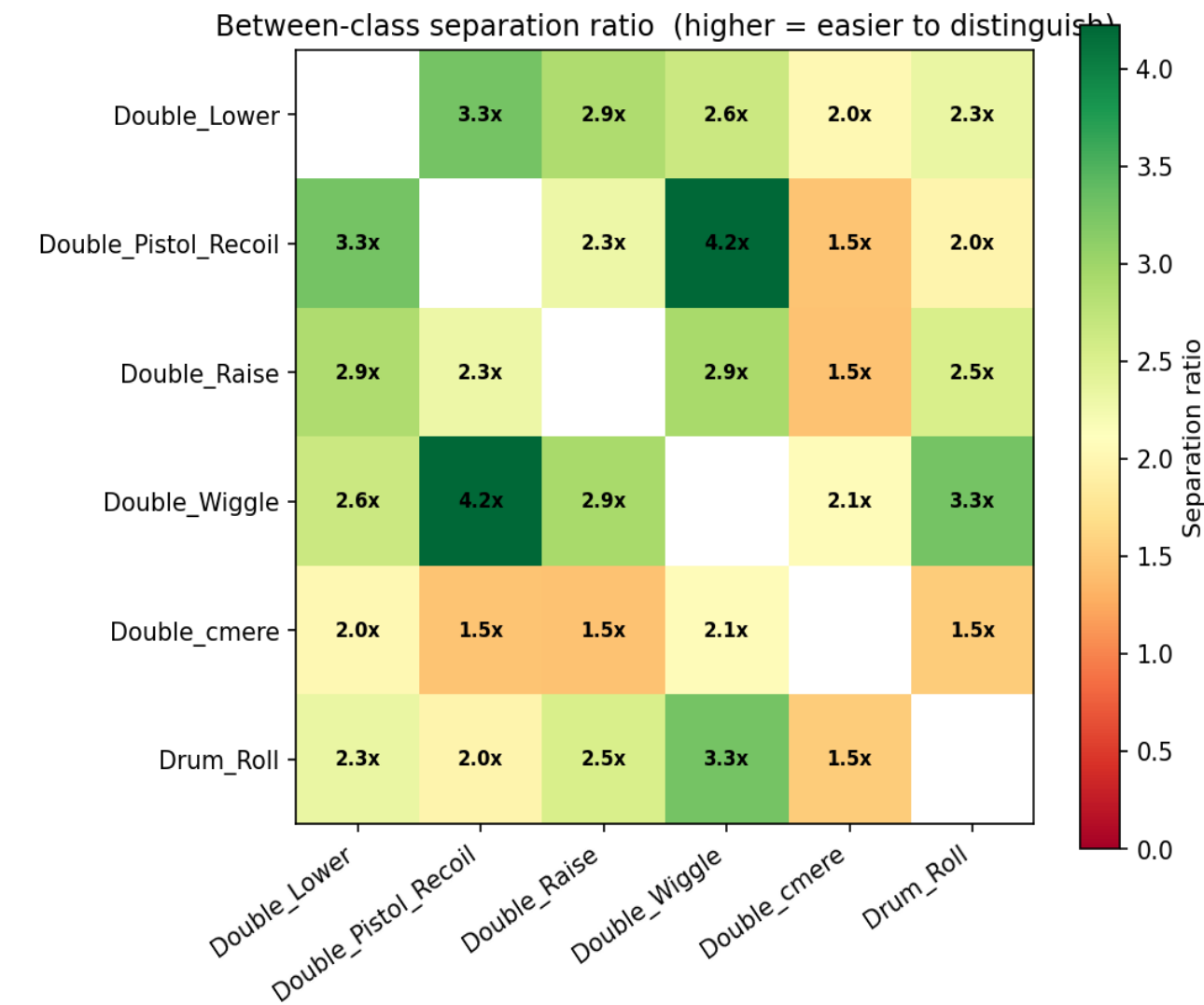
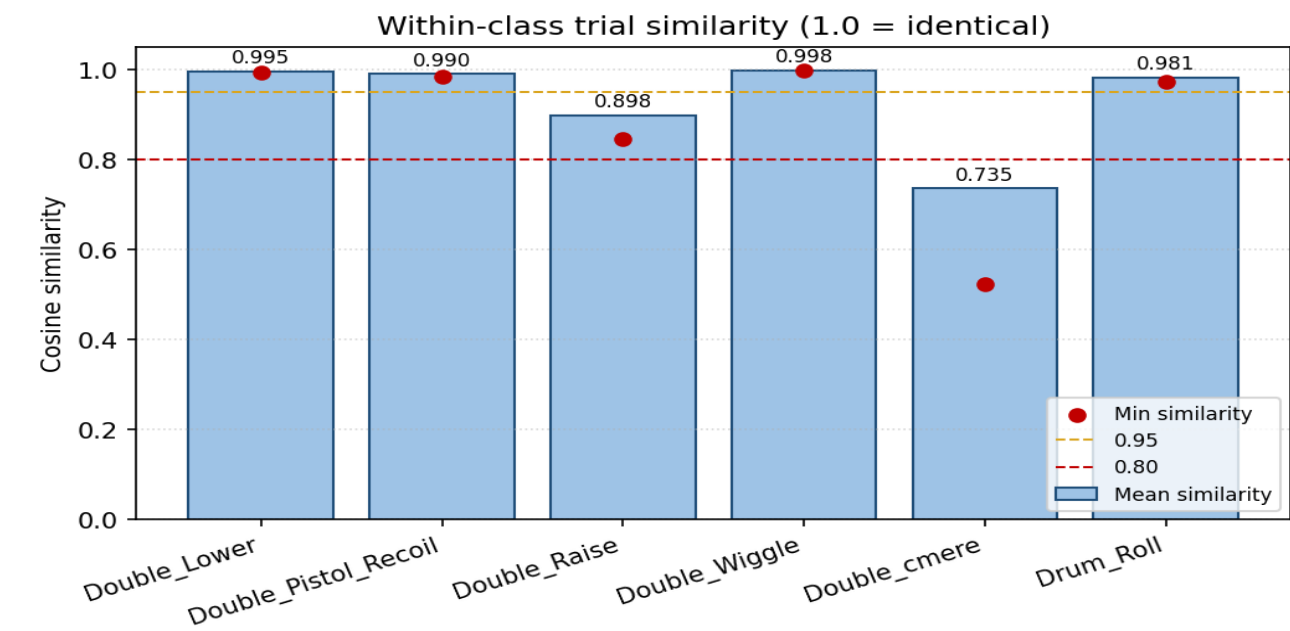


t-SNE feature space projection

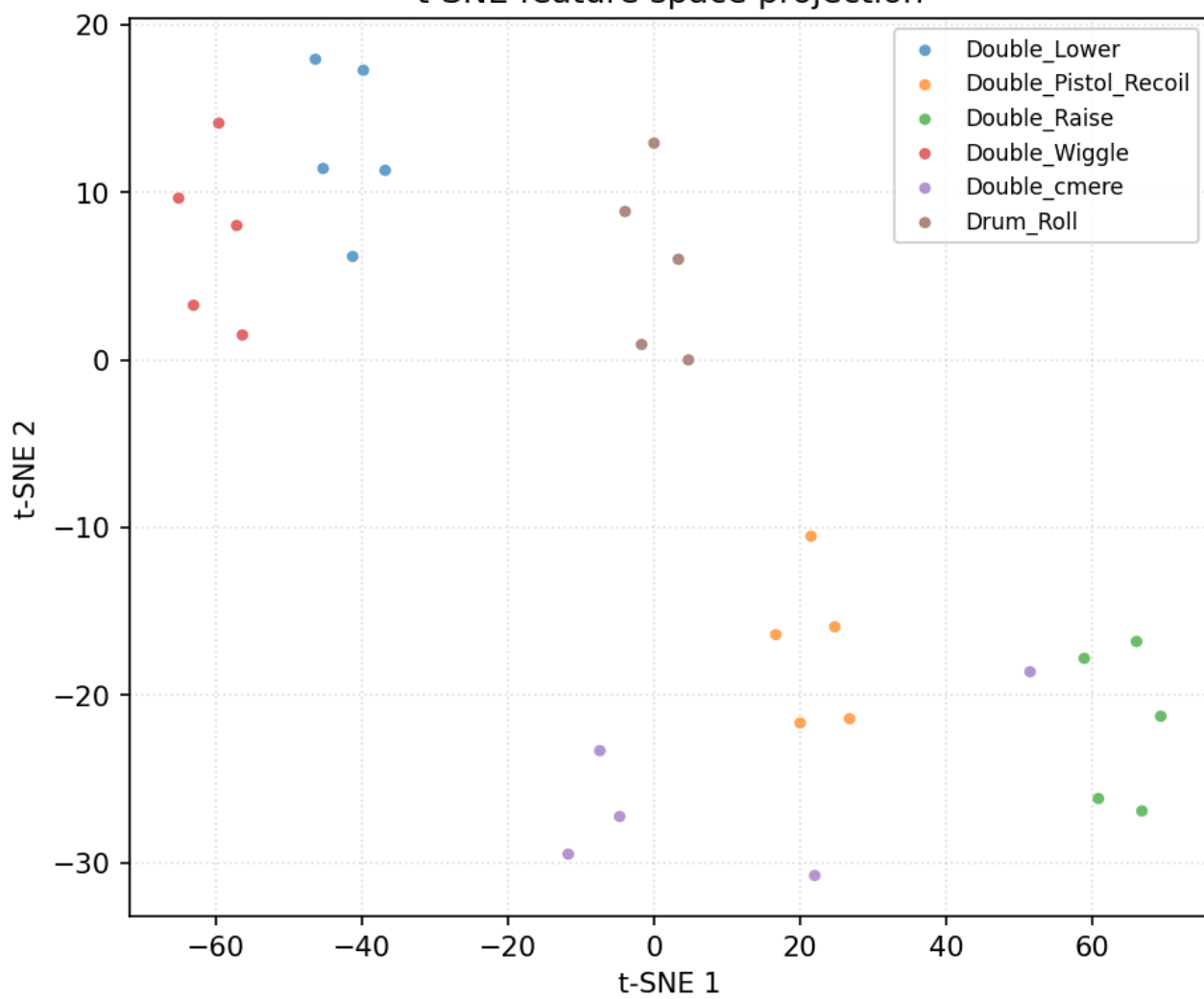


Data diagnostics — user 14_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

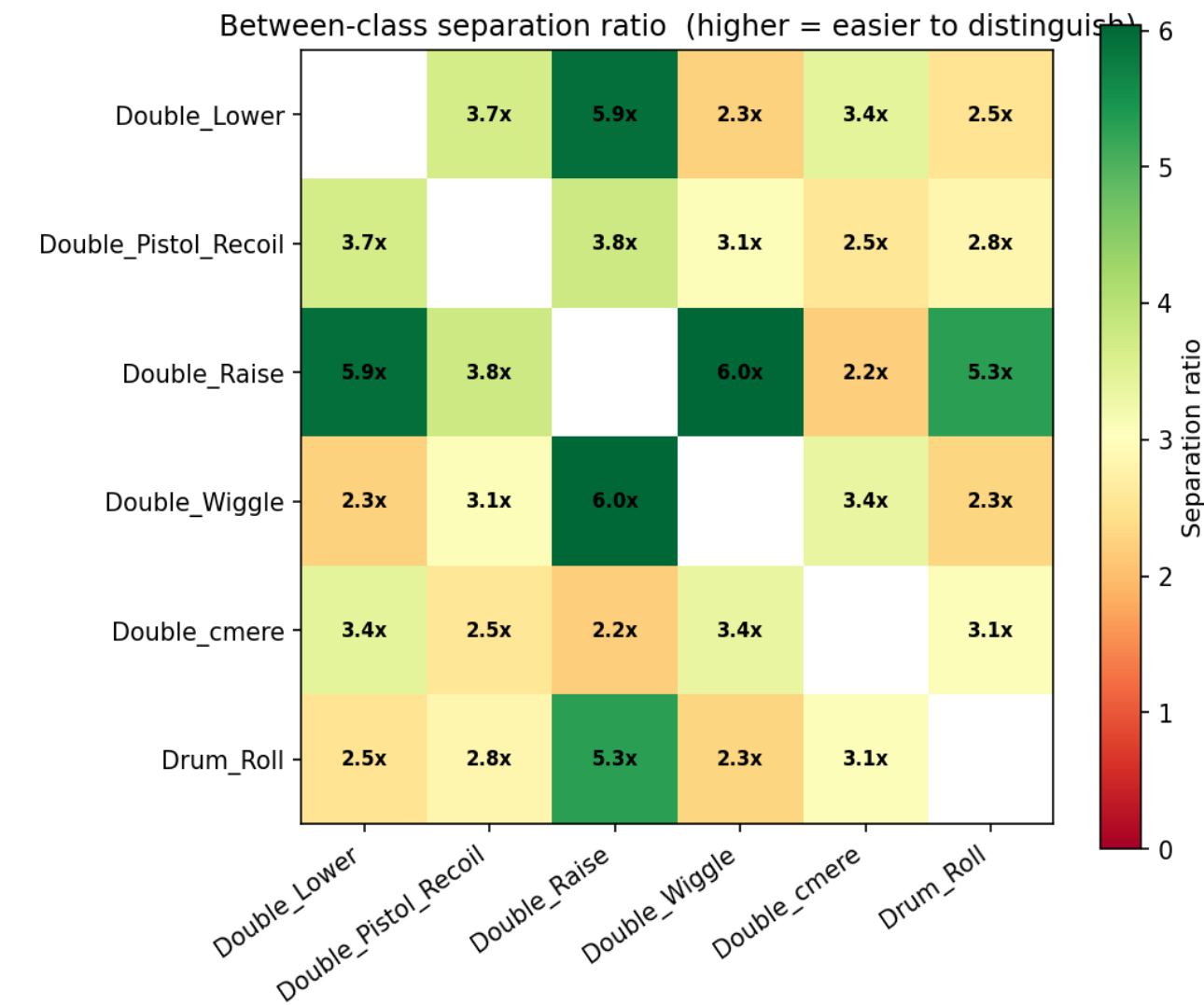
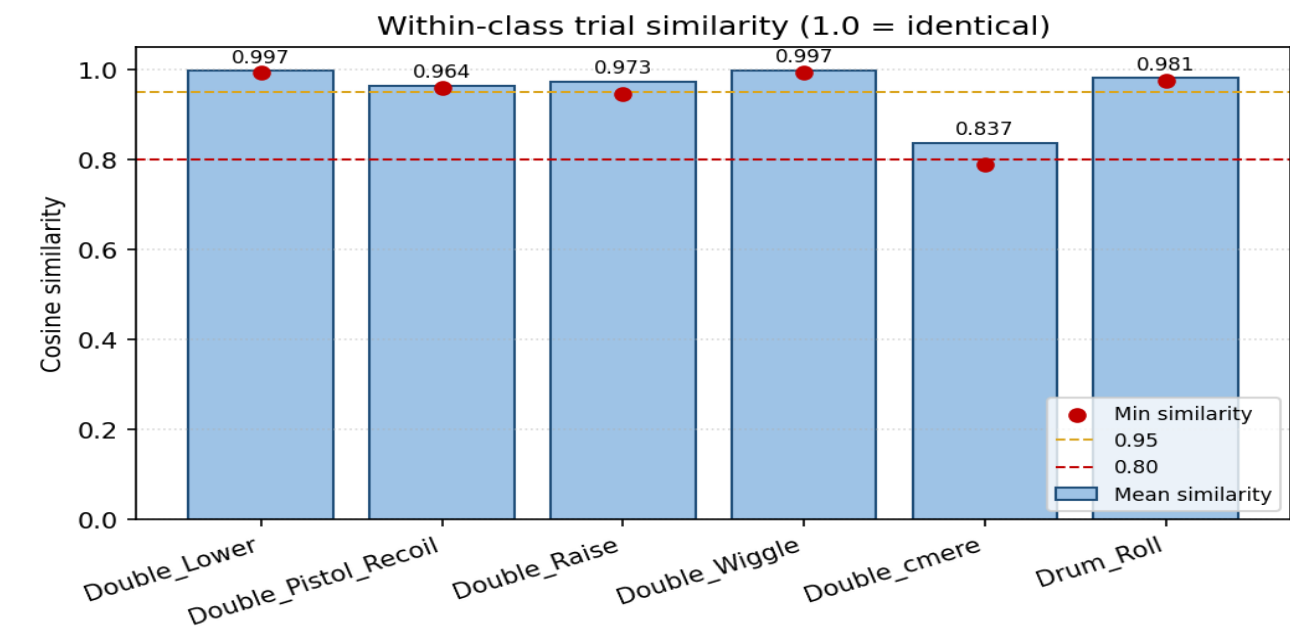


t-SNE feature space projection

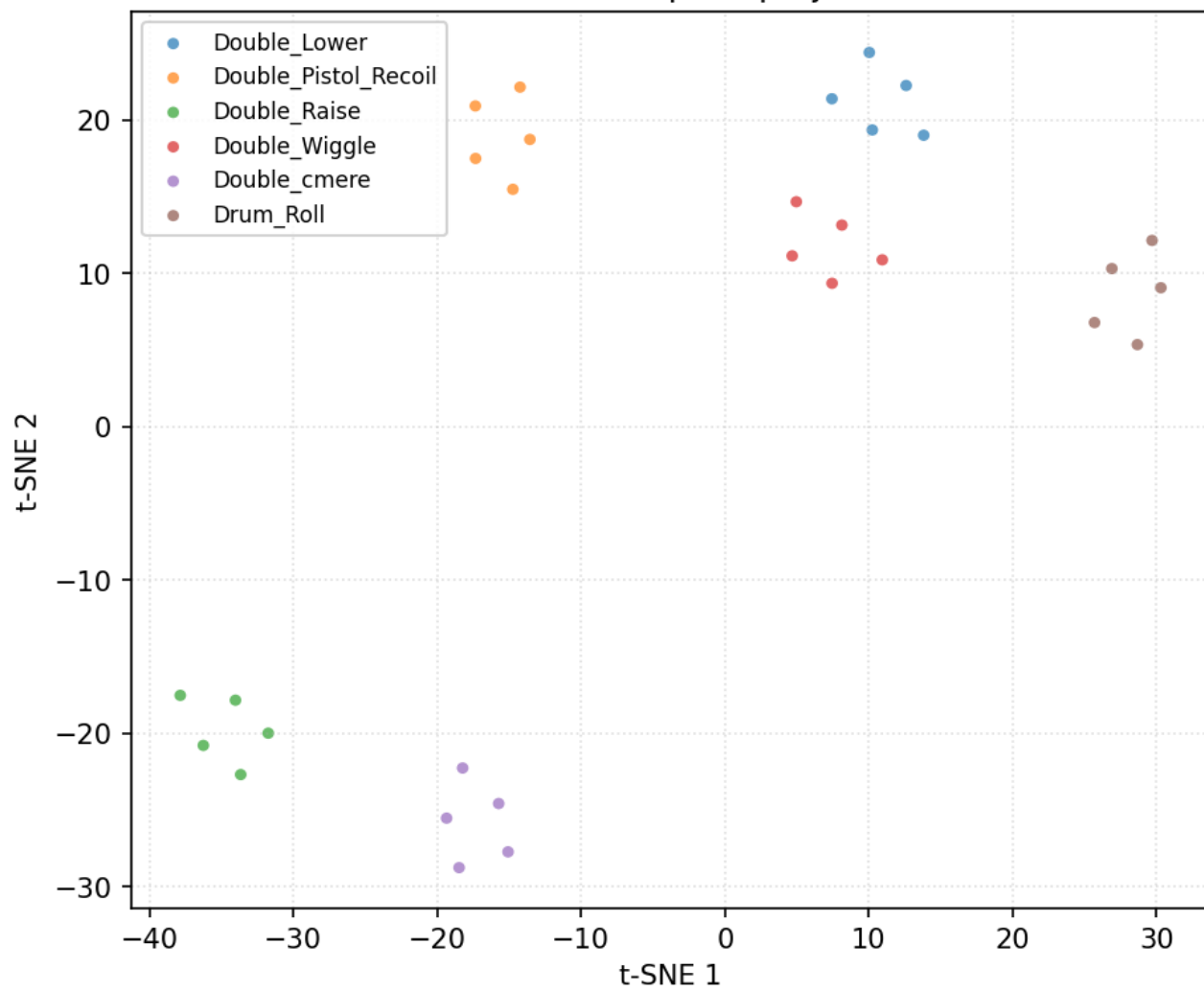


Data diagnostics — user 17_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

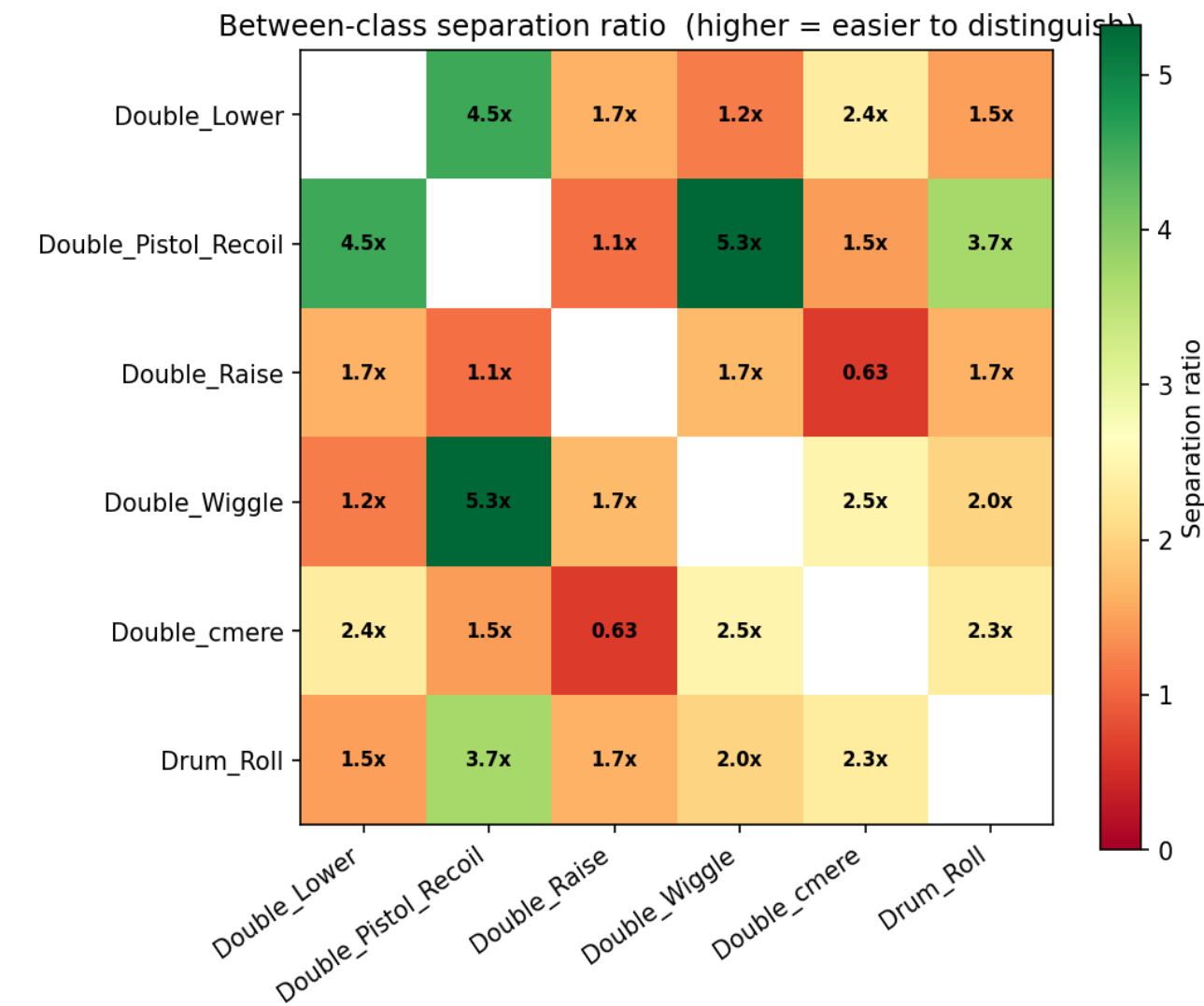
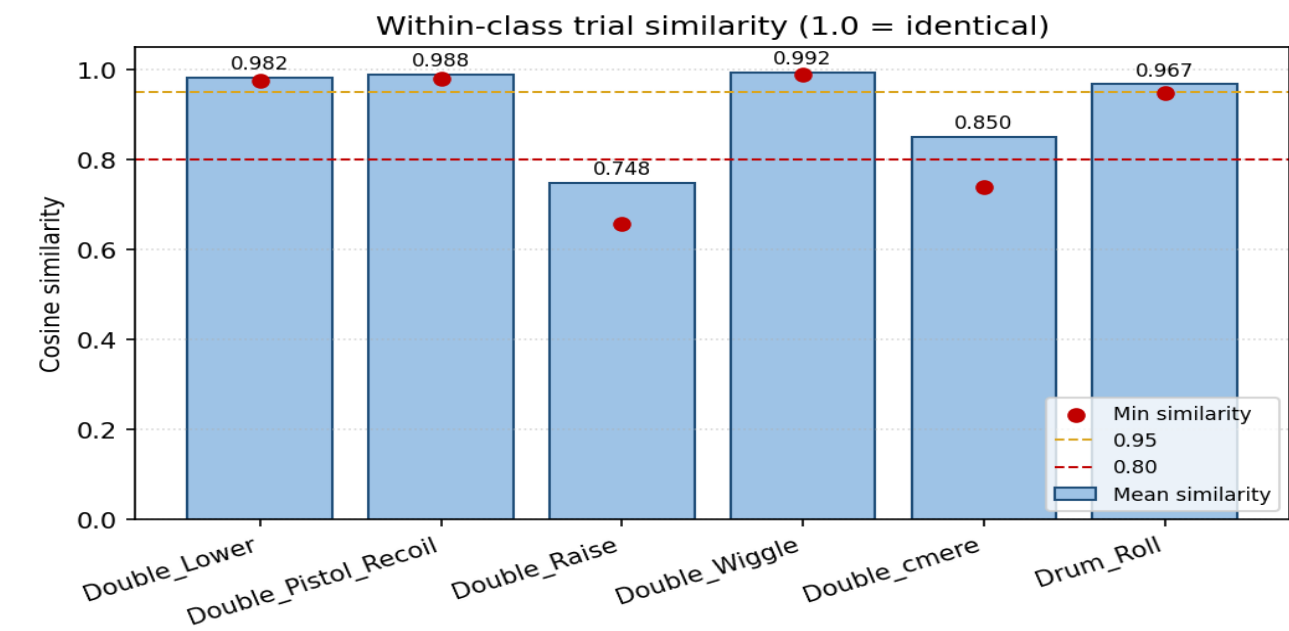


t-SNE feature space projection

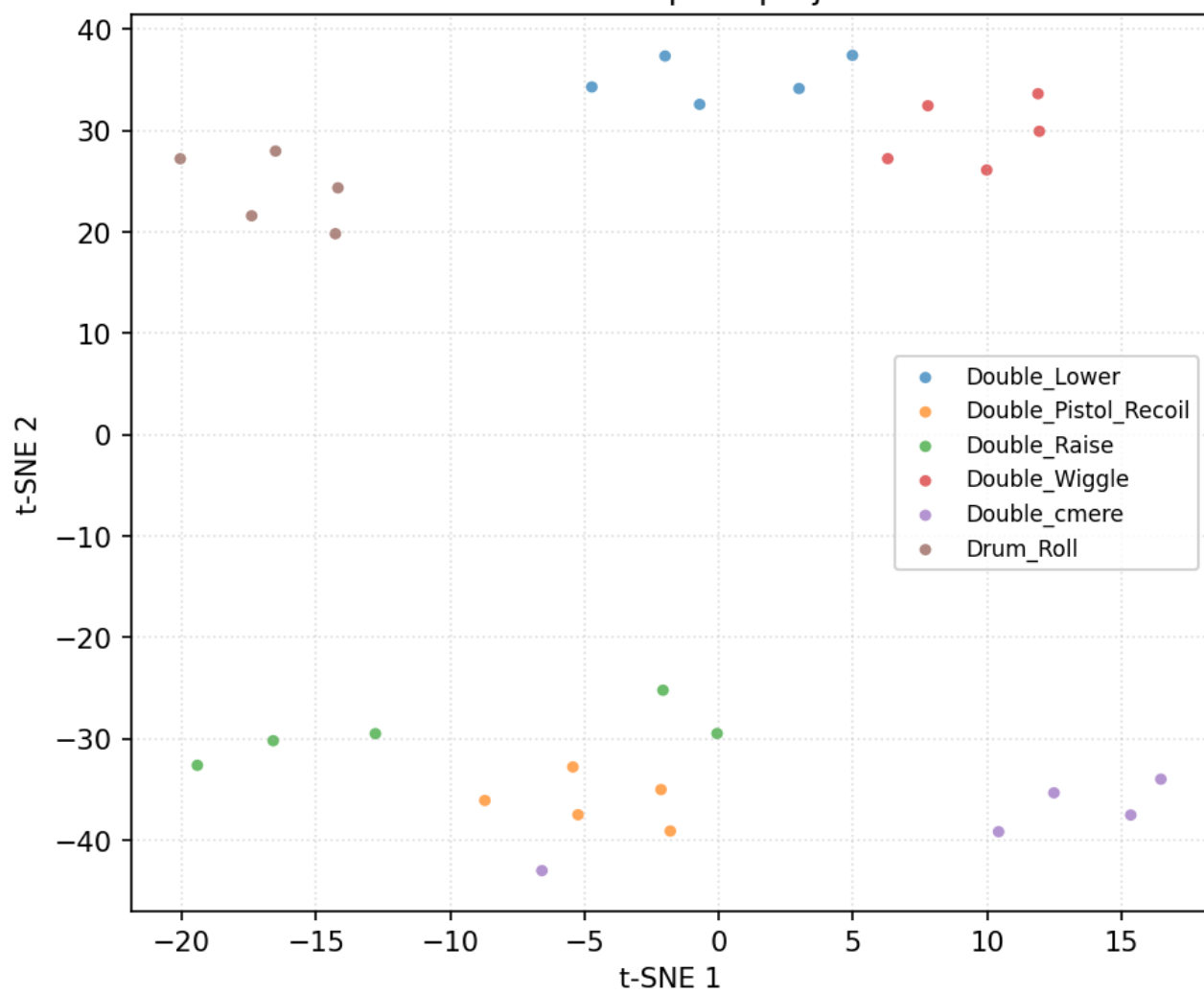


Data diagnostics — user 18_Bridgette

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

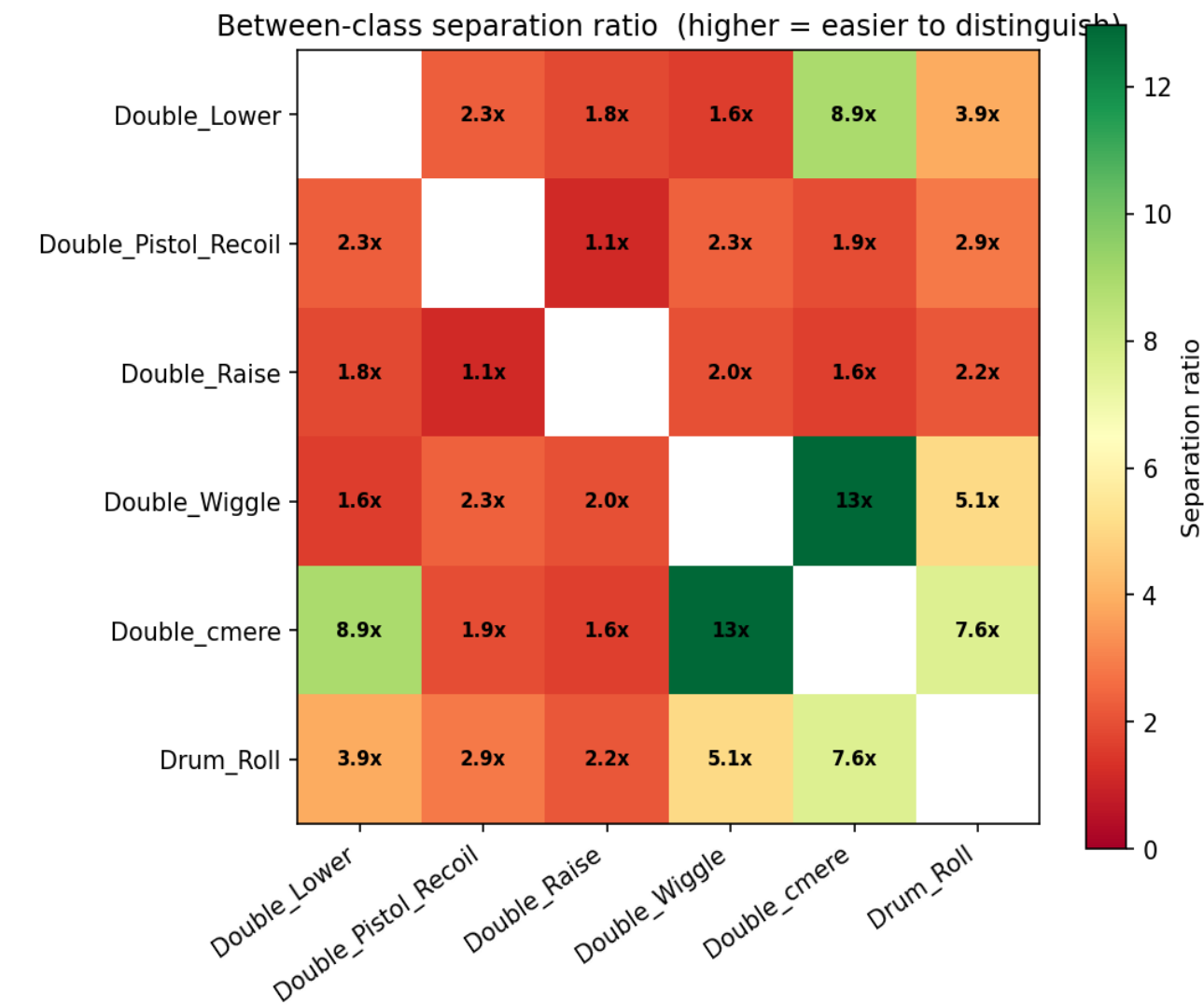
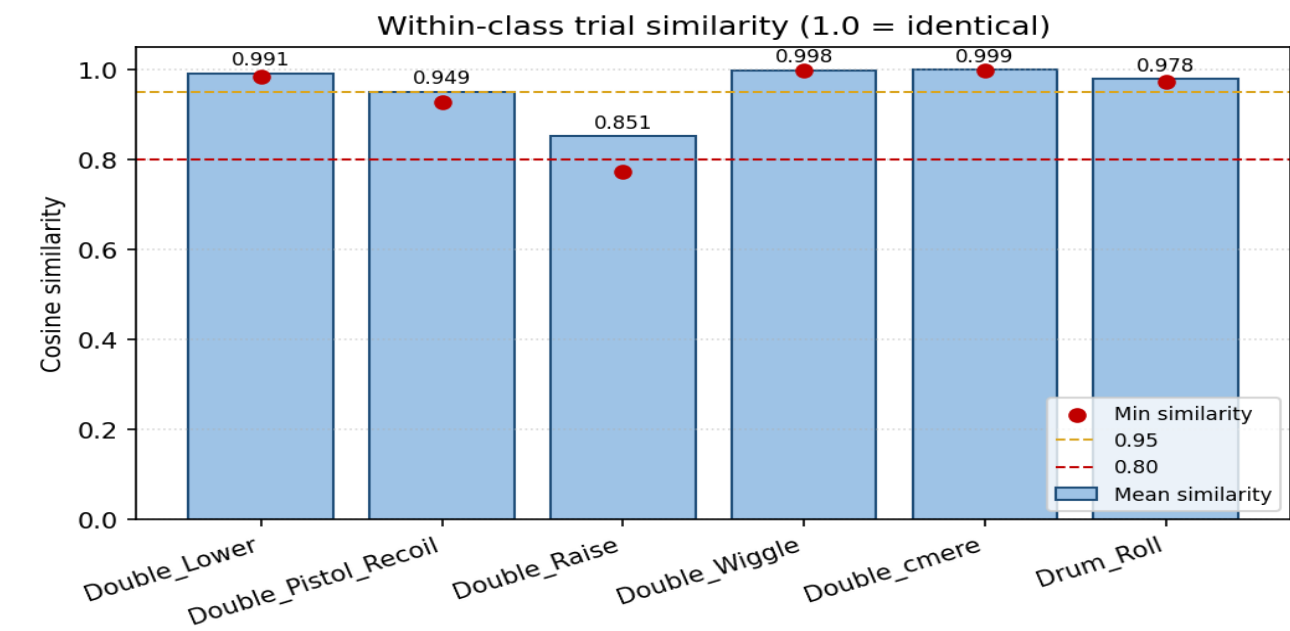


t-SNE feature space projection

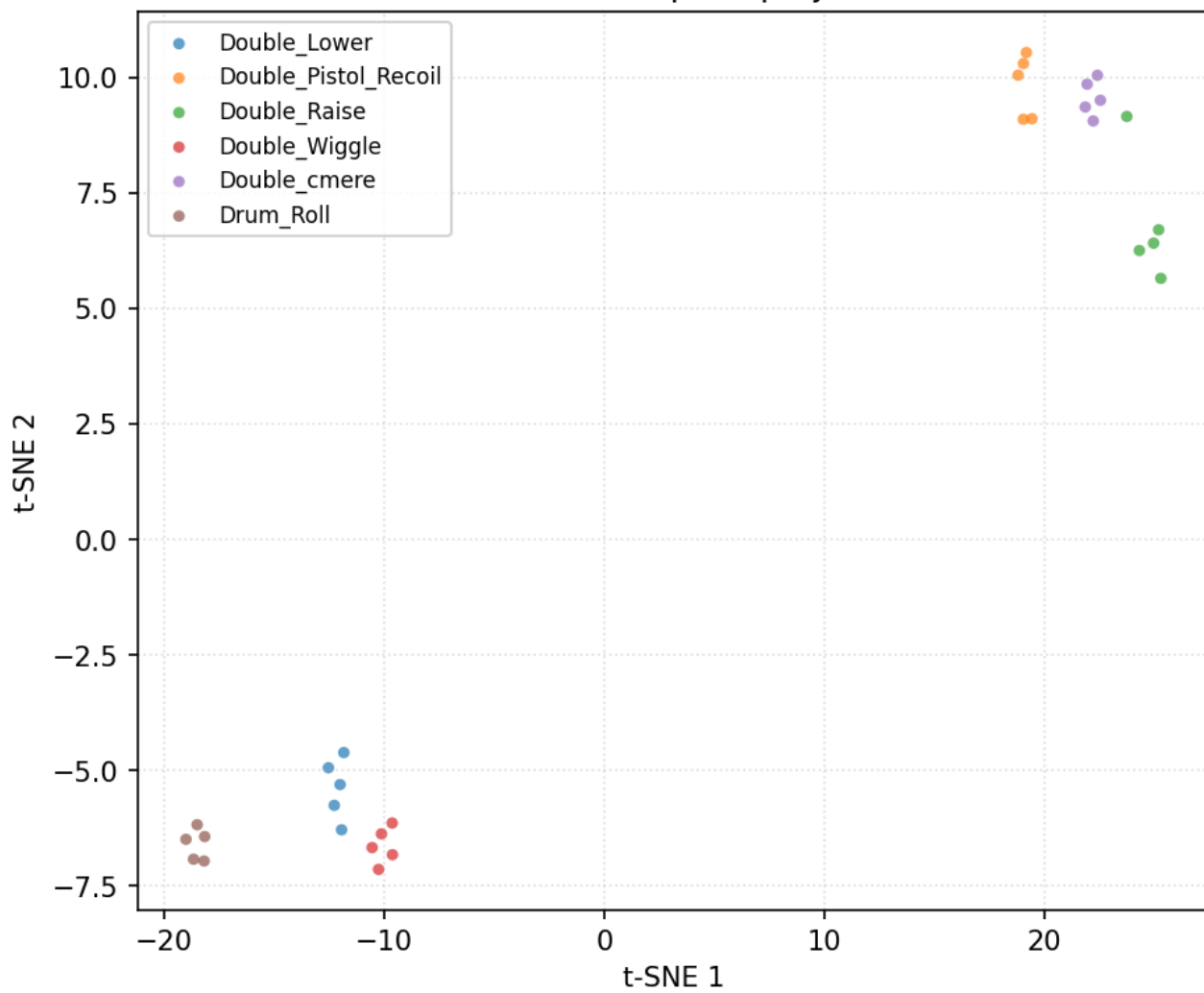


Data diagnostics — user 1_Alán

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

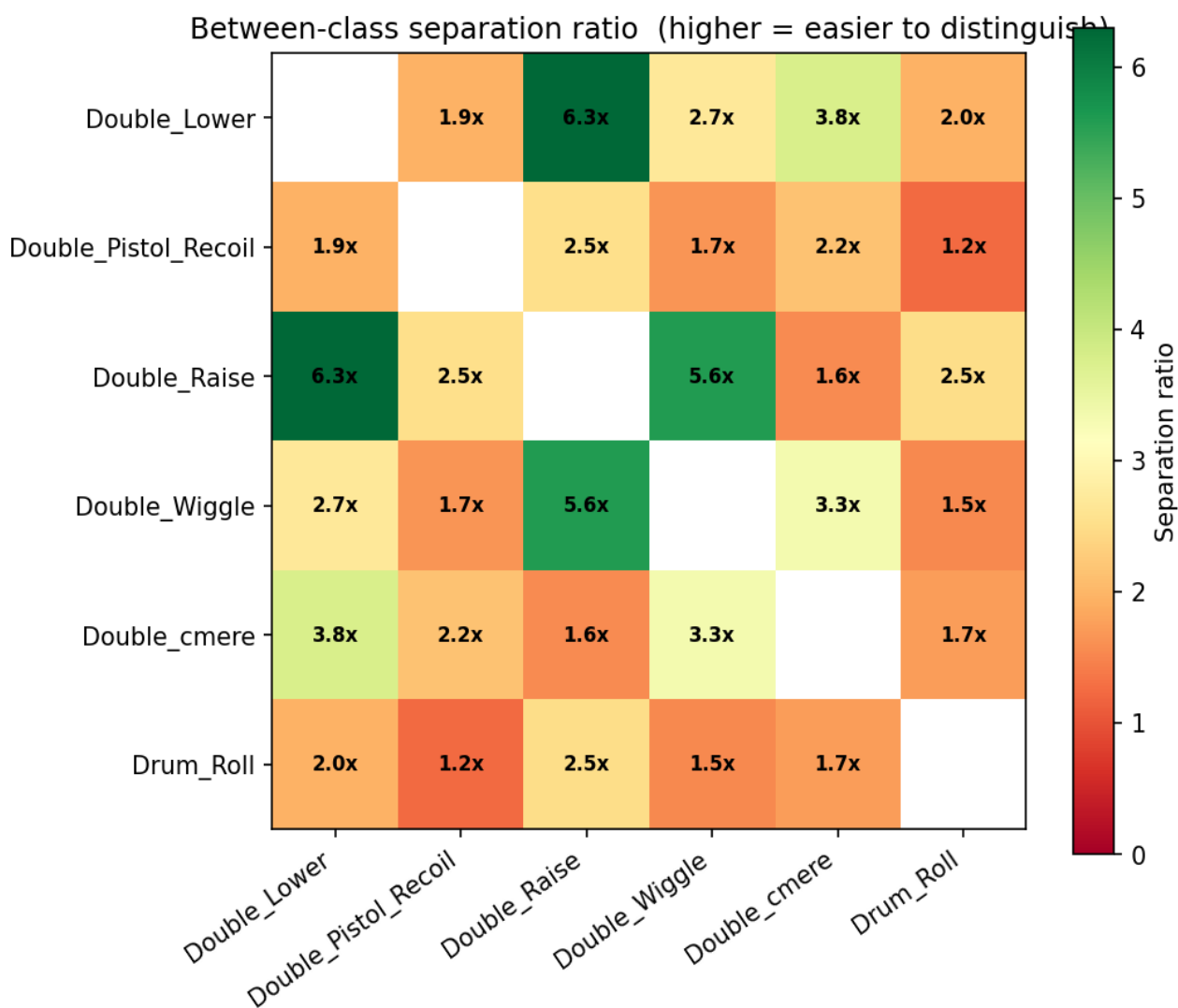
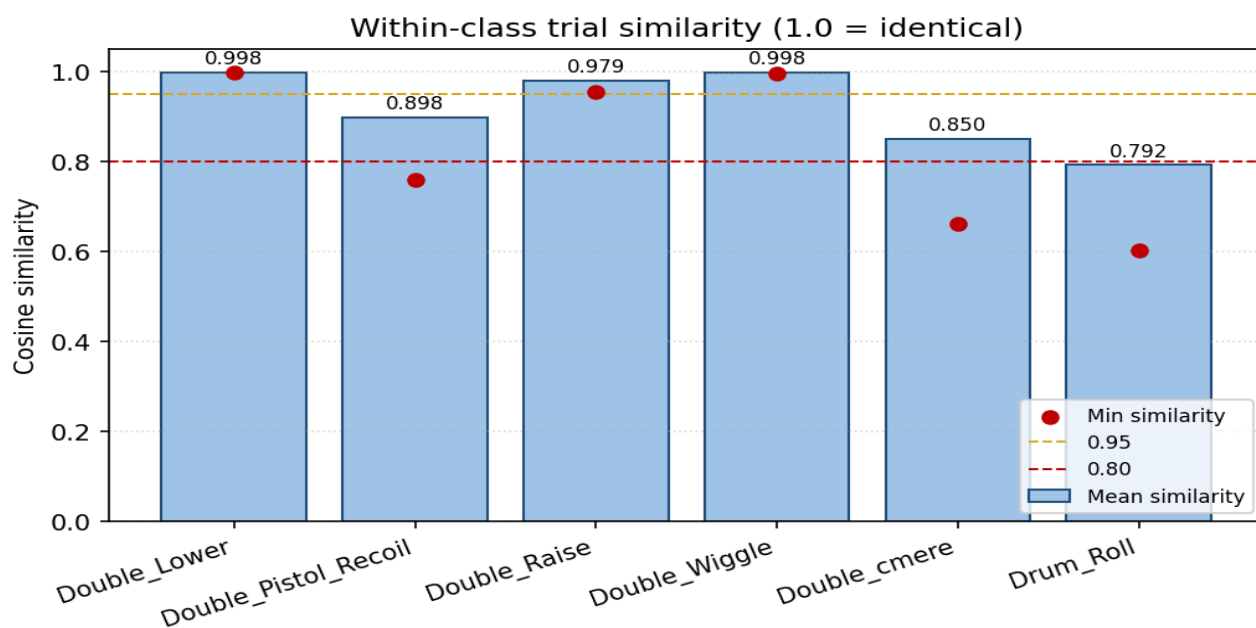


t-SNE feature space projection

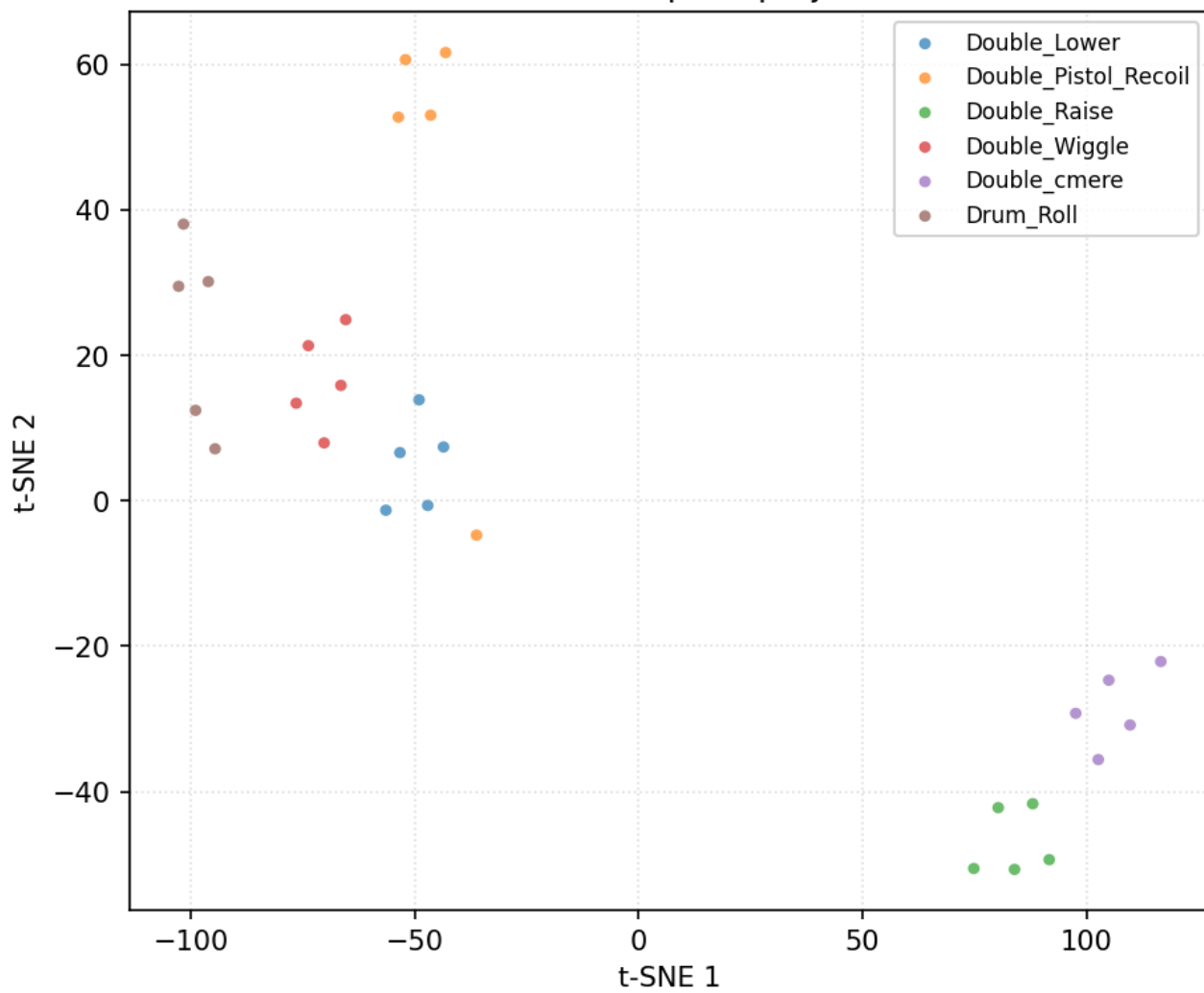


Data diagnostics — user 2_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

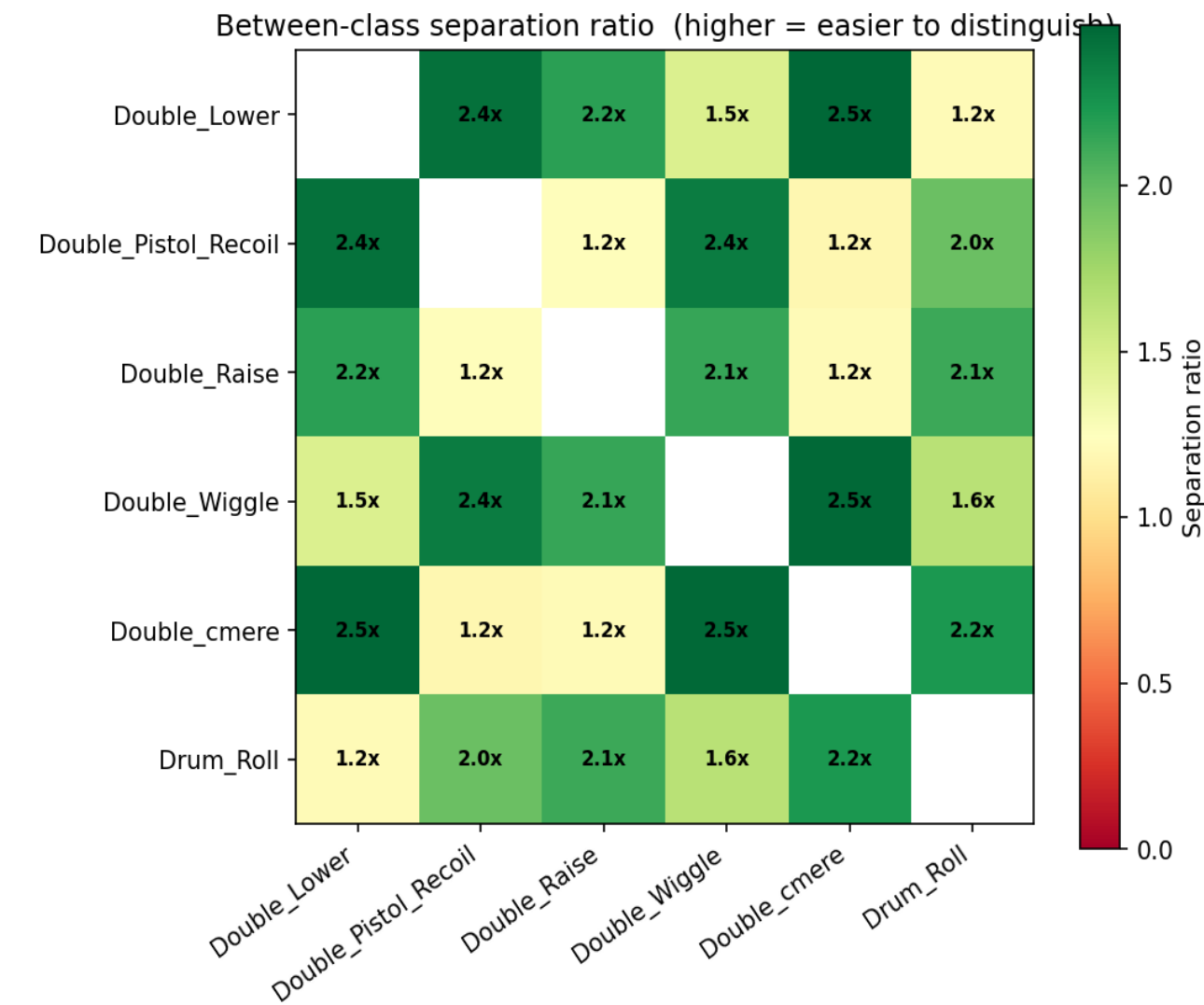
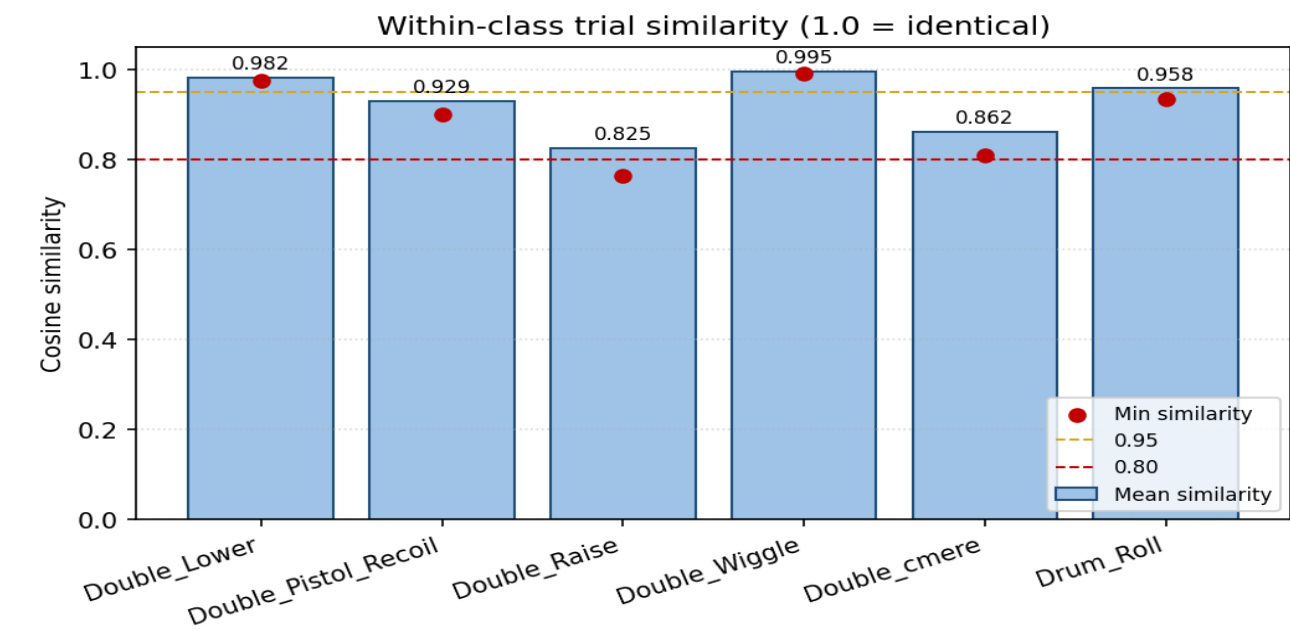


t-SNE feature space projection

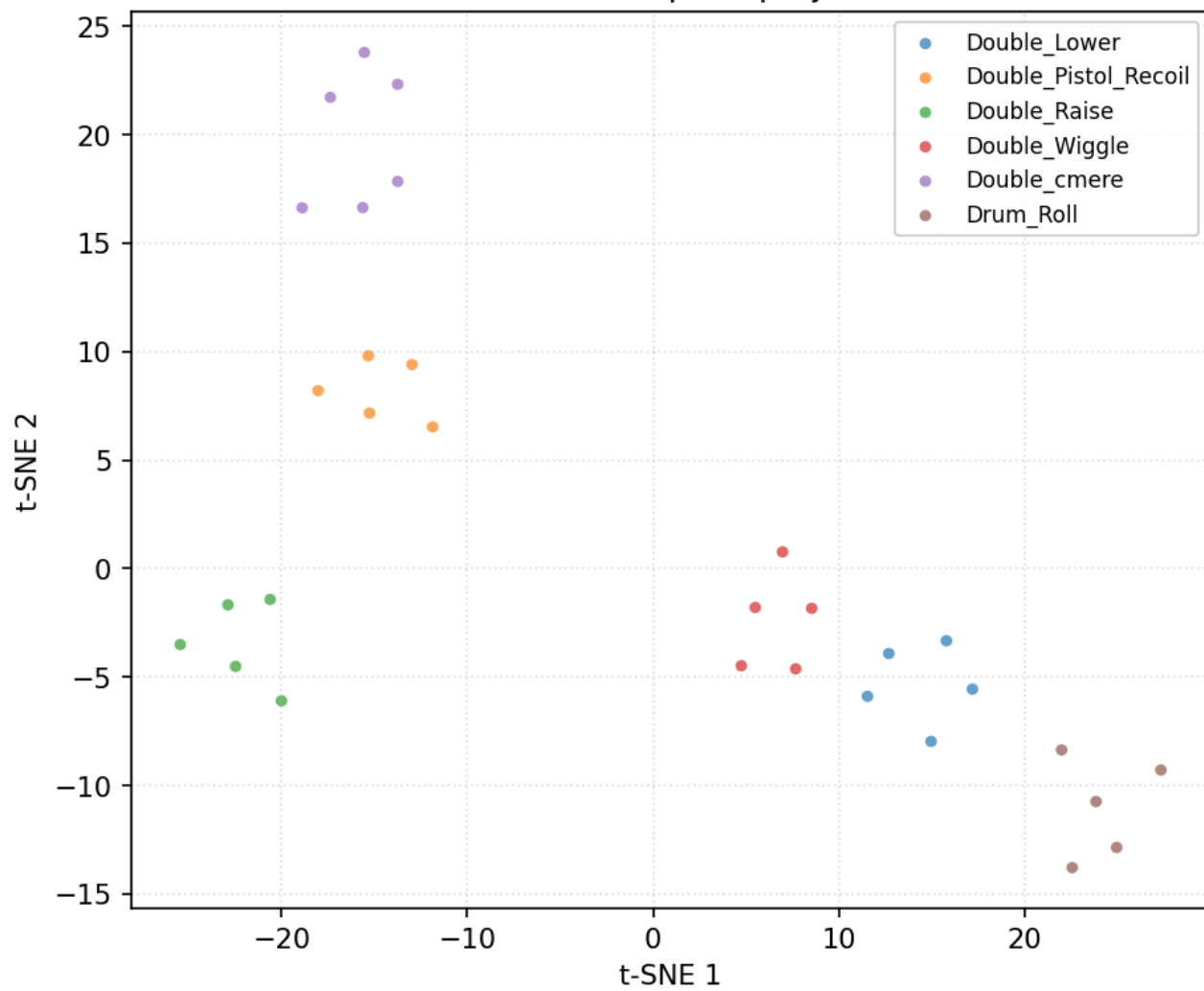


Data diagnostics — user 3_Anghad

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

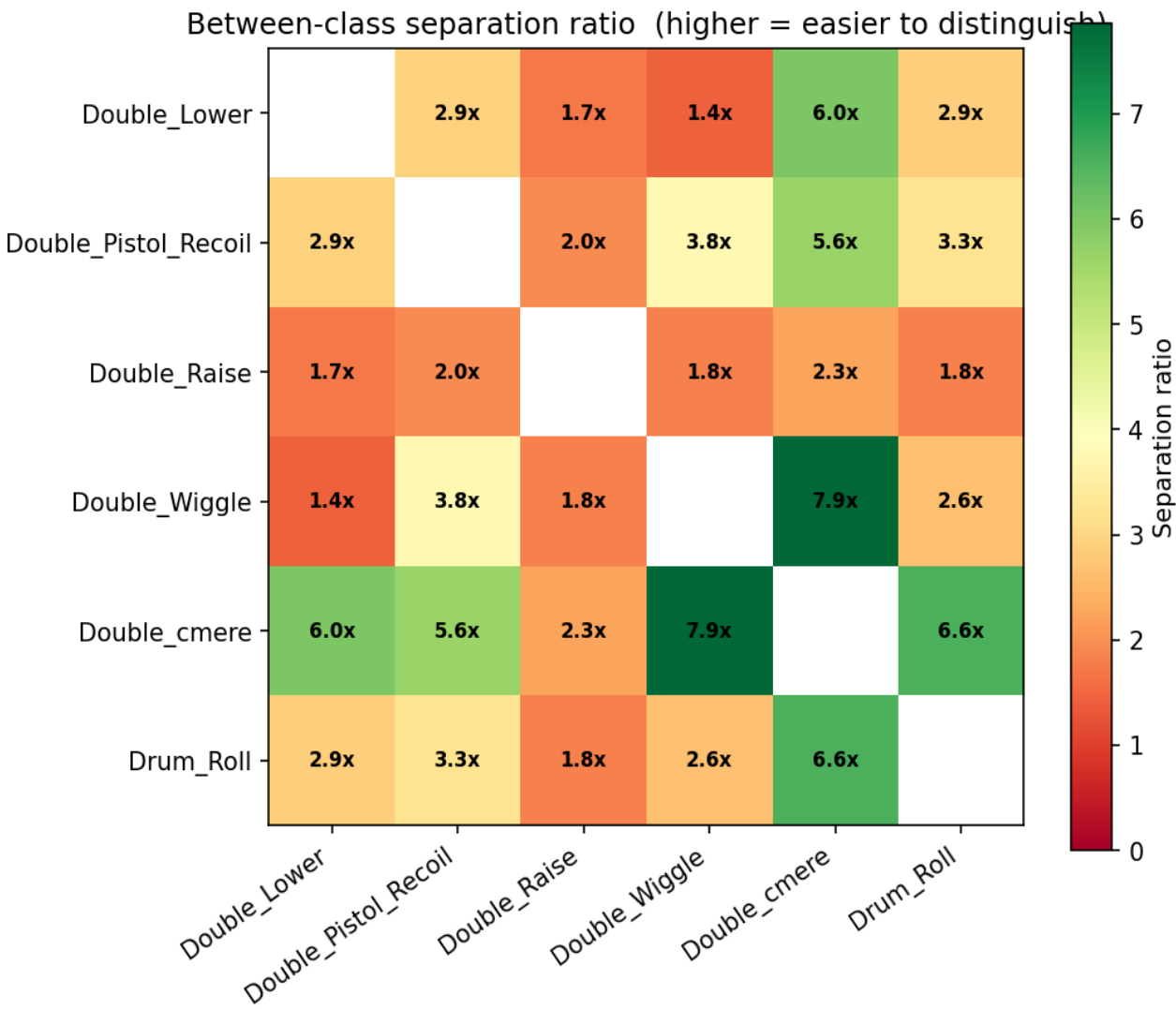
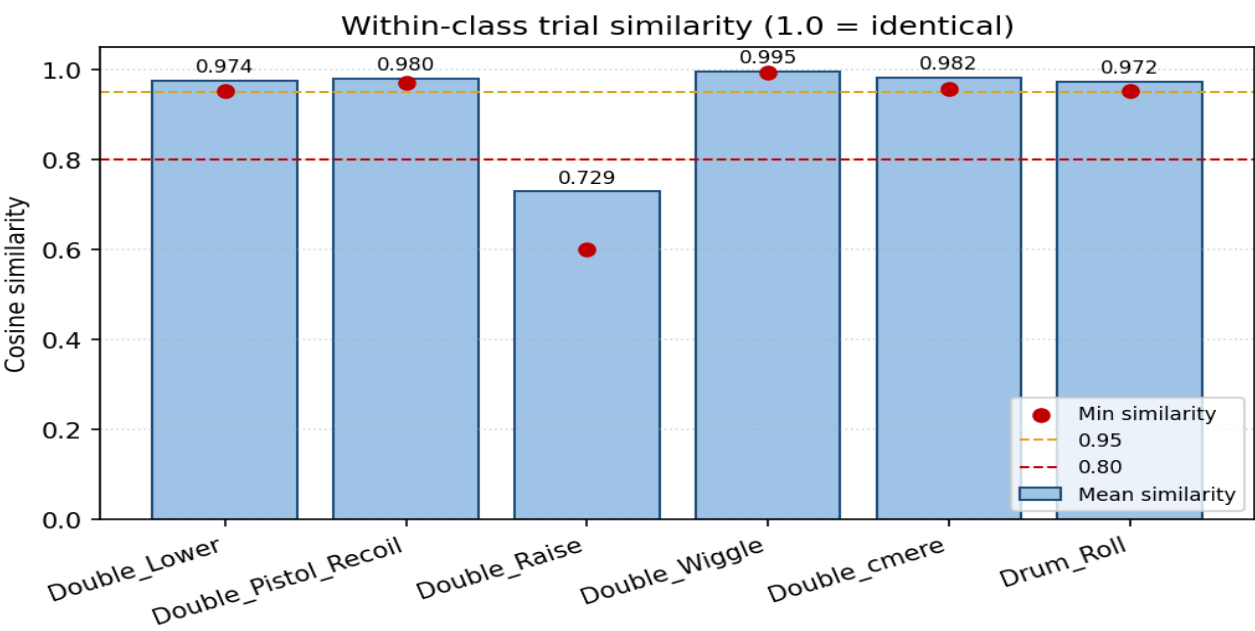


t-SNE feature space projection

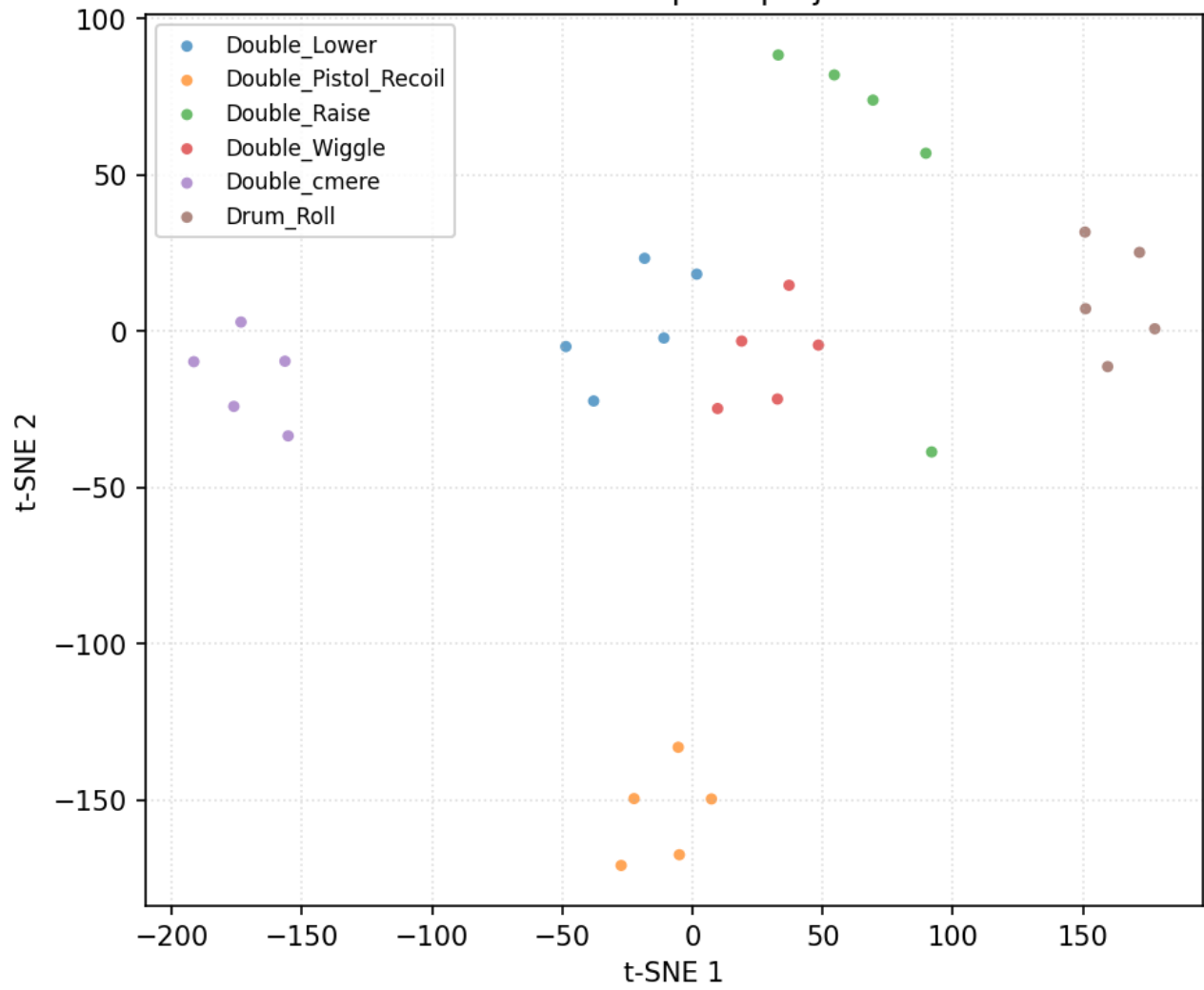


Data diagnostics — user 4_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

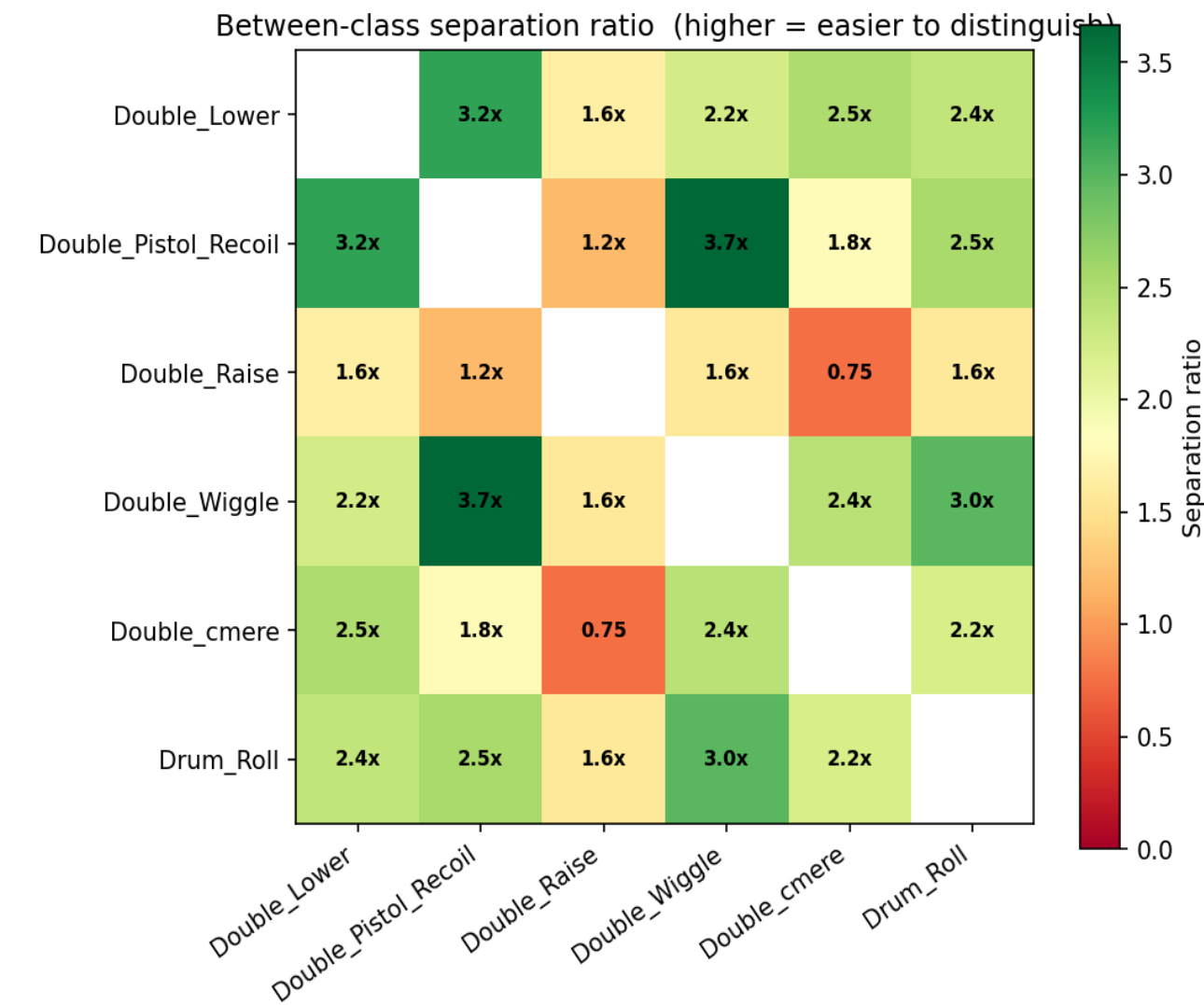
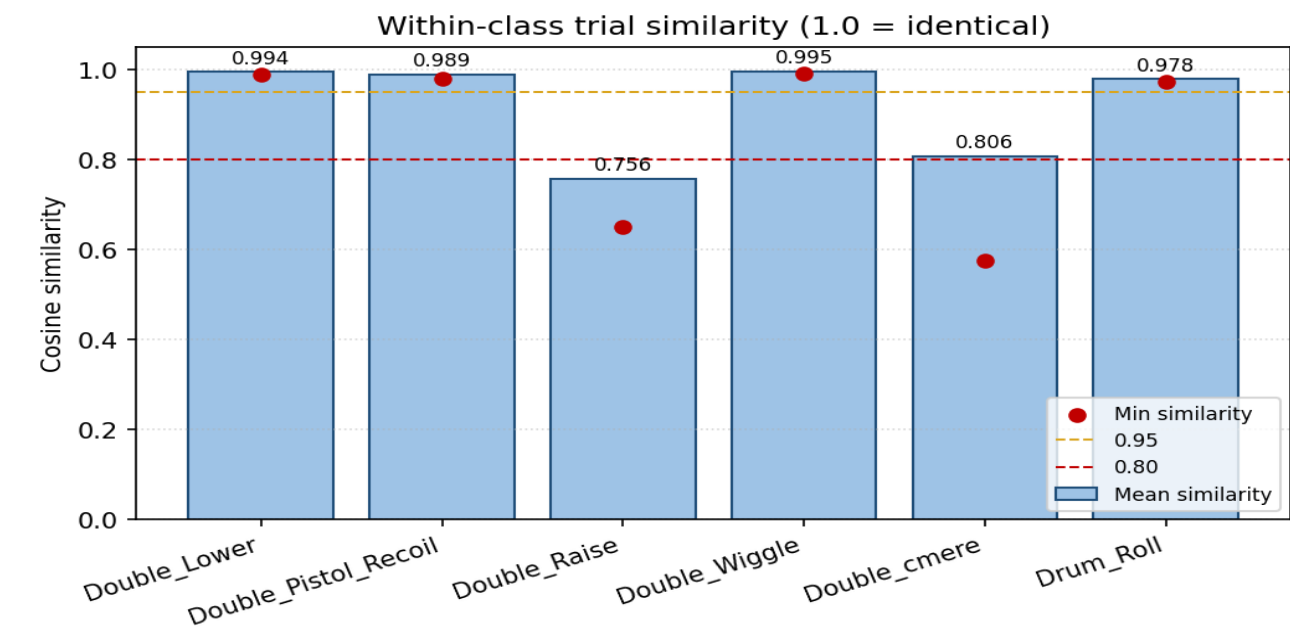


t-SNE feature space projection

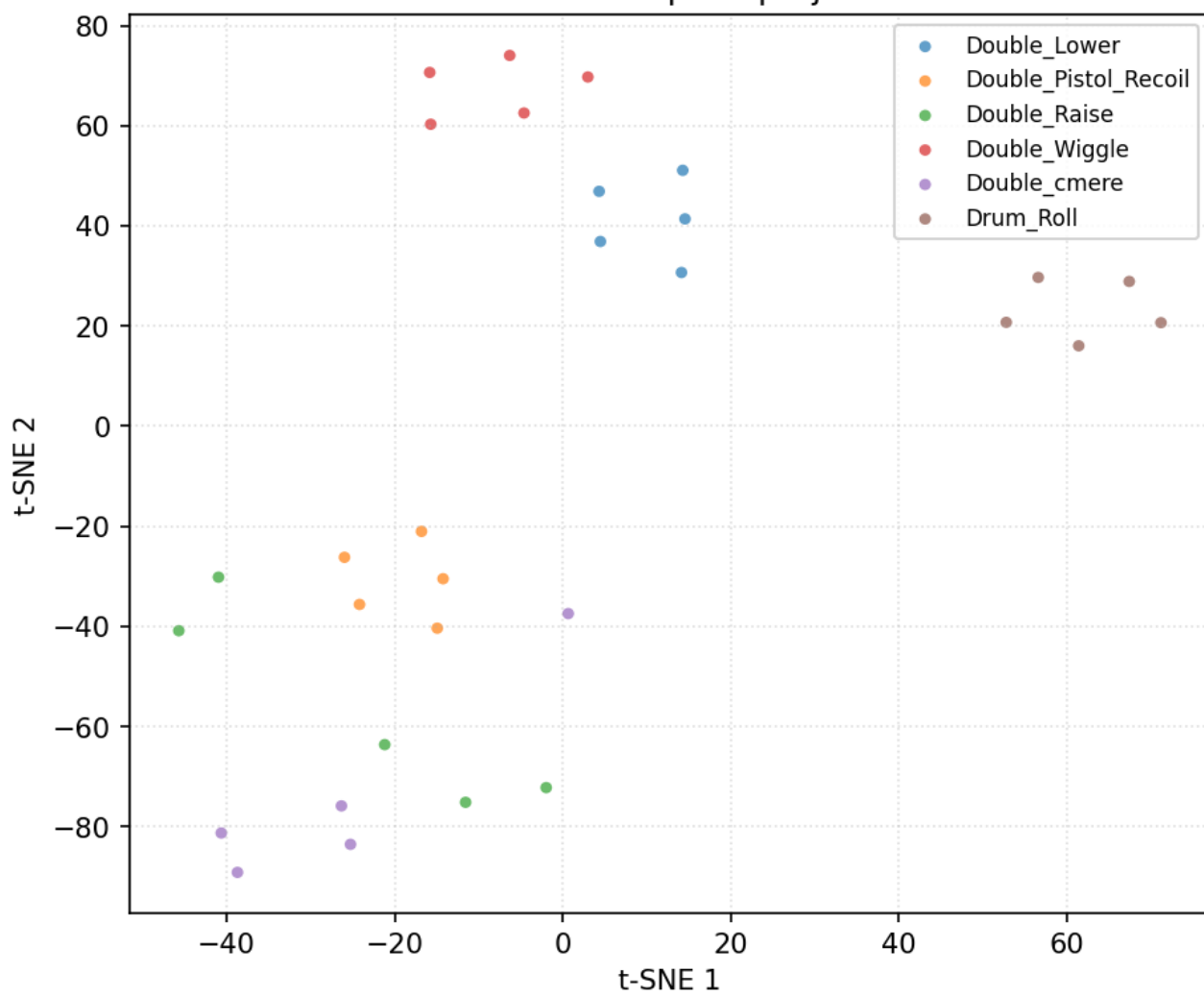


Data diagnostics — user 6_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

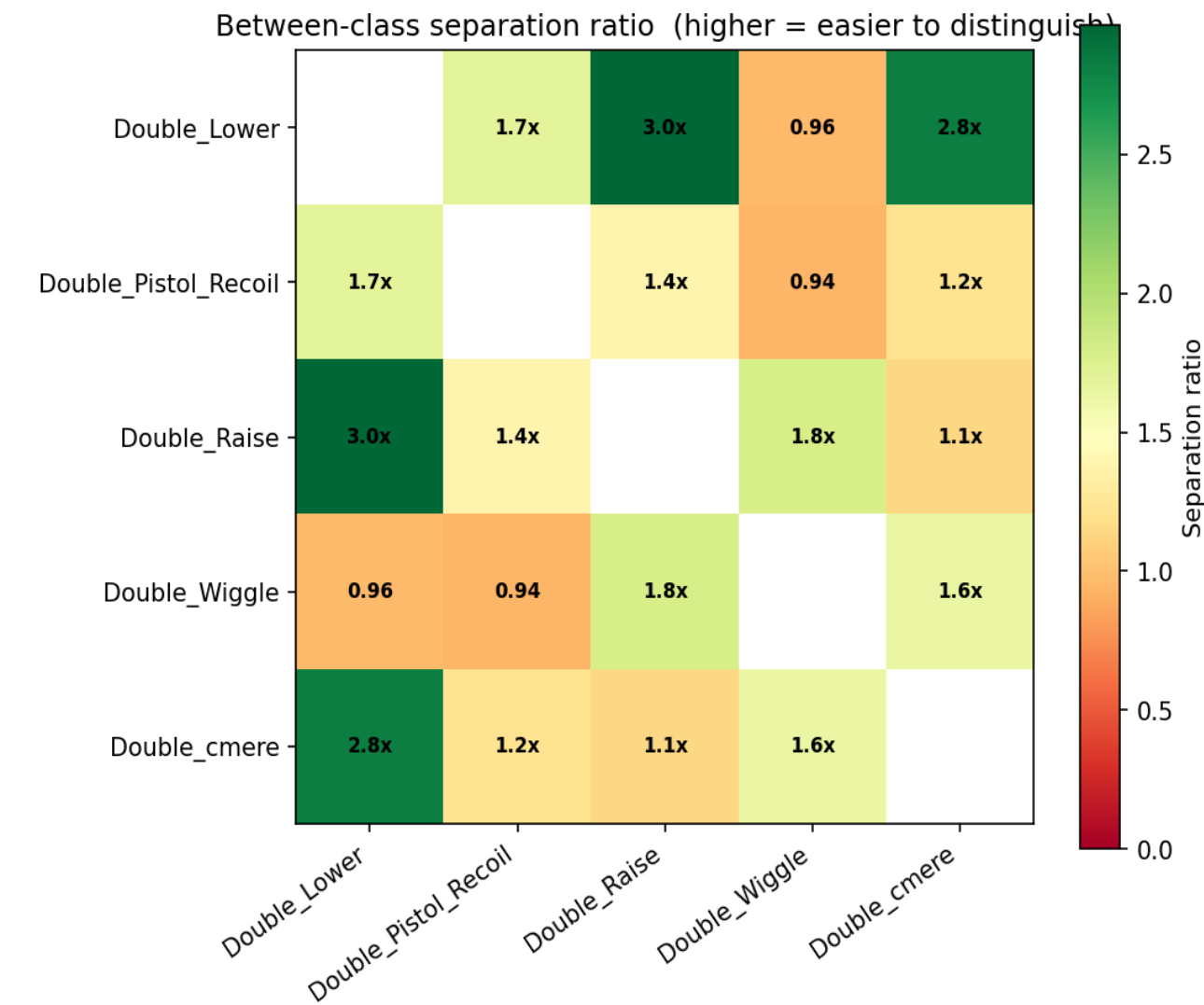
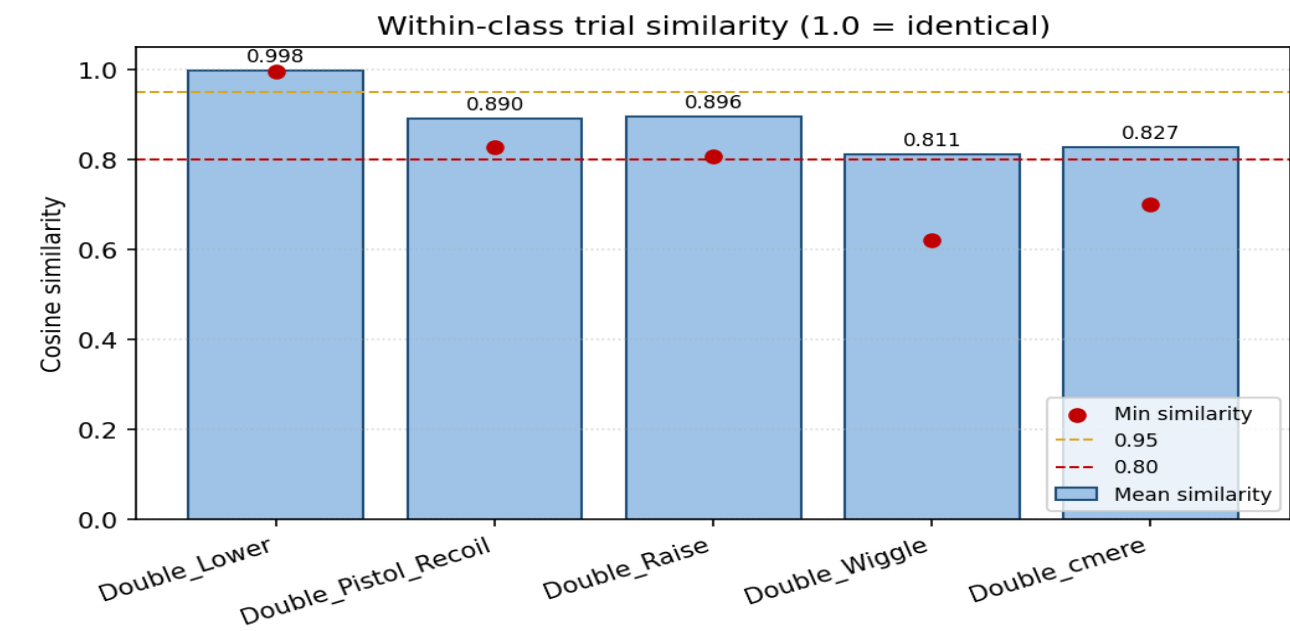


t-SNE feature space projection

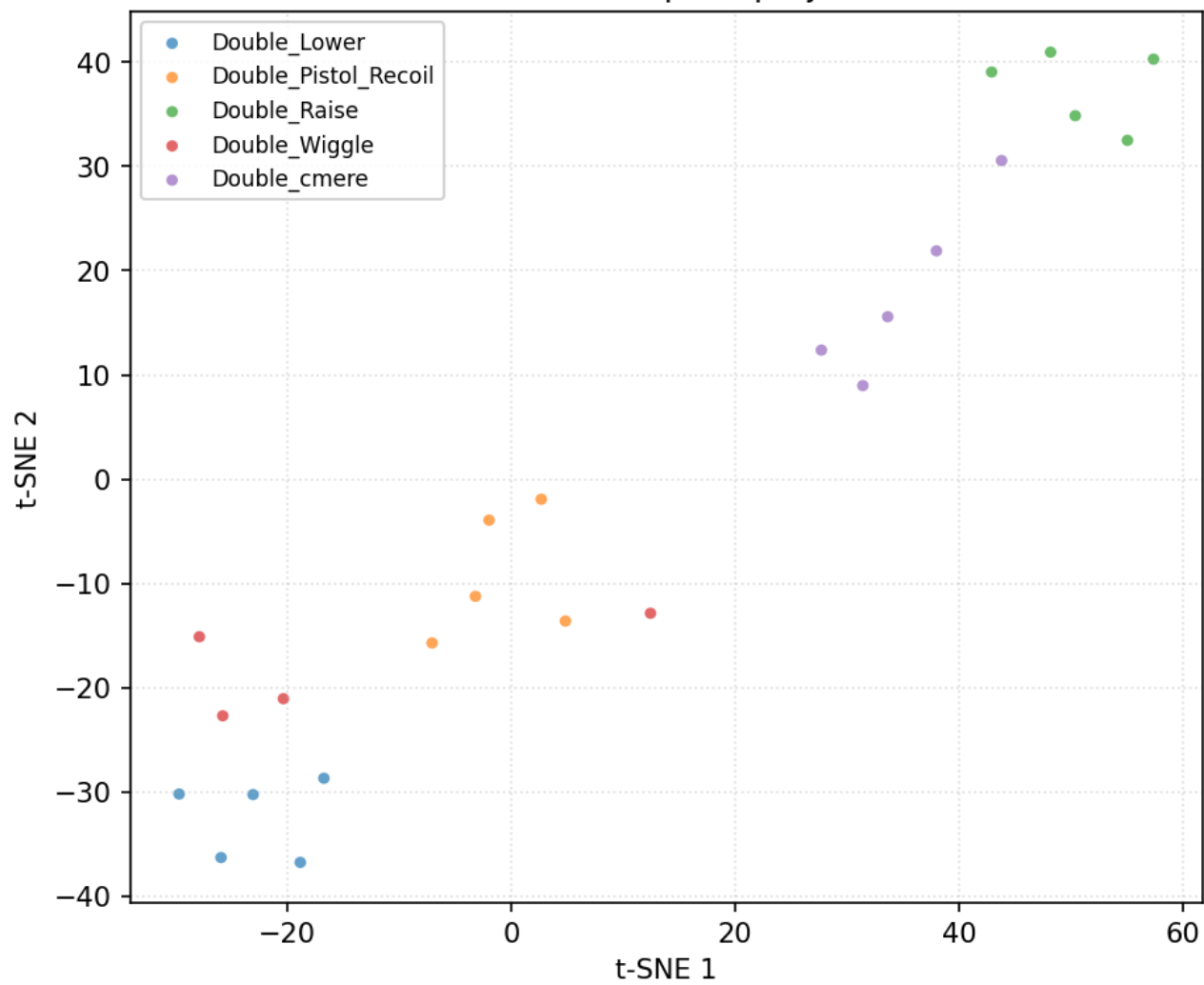


Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	36,966
LOUO mean ± std	0.9127 ± 0.0630
Folds run	11
Inner-val mode	user
Per-trial norm	zscore

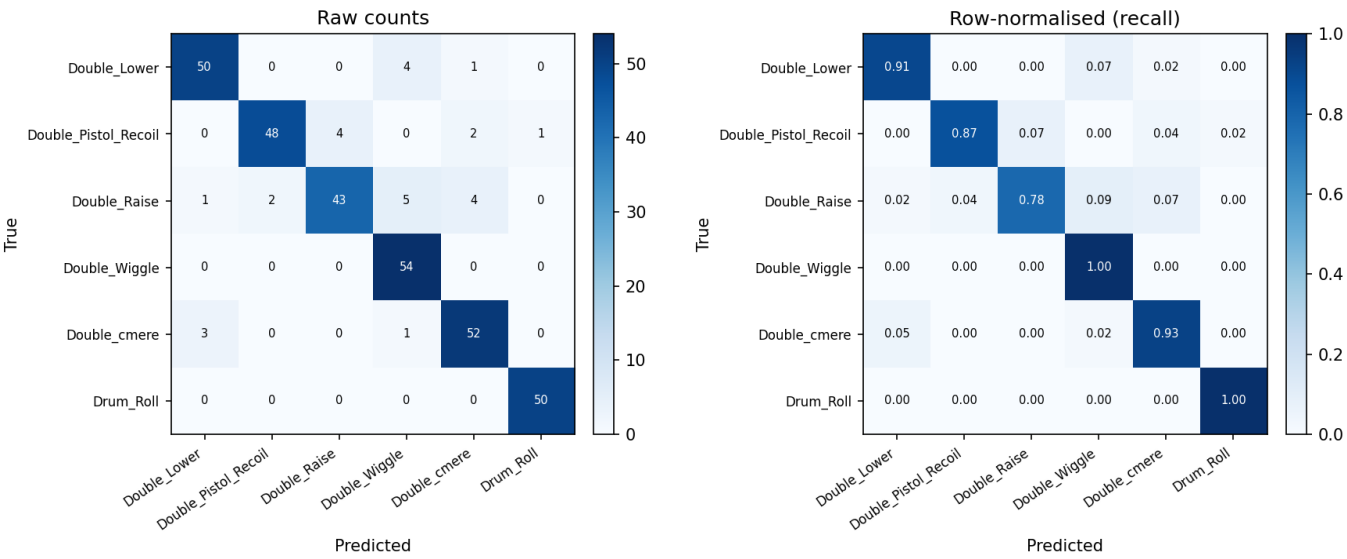
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	795	30	0.8667
2	12_Marcus	14_StephenV2	795	30	1.0000
3	14_StephenV2	17_Raquel	795	30	0.9000
4	17_Raquel	18_Bridgette	795	30	0.9667
5	18_Bridgette	1_Alan	795	30	0.9000
6	1_Alan	2_Alex	795	30	1.0000
7	2_Alex	3_Anghad	792	30	0.9667
8	3_Anghad	4_Daniel	792	31	0.8065
9	4_Daniel	6_Harry	795	30	0.8667
10	6_Harry	8_Jestin	813	30	0.9333
11	8_Jestin	11_Mansh	813	24	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.926	0.909	0.917	55
Double_Pistol_Recoil	0.960	0.873	0.914	55
Double_Raise	0.915	0.782	0.843	55
Double_Wiggle	0.844	1.000	0.915	54
Double_cmere	0.881	0.929	0.904	56
Drum_Roll	0.980	1.000	0.990	50
macro avg	0.918	0.915	0.914	325
weighted avg	0.917	0.914	0.913	325

Confusion matrix (aggregated across folds)



Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	183,654
LOUO mean ± std	0.8575 ± 0.0602
Folds run	11
Inner-val mode	user
Per-trial norm	zscore

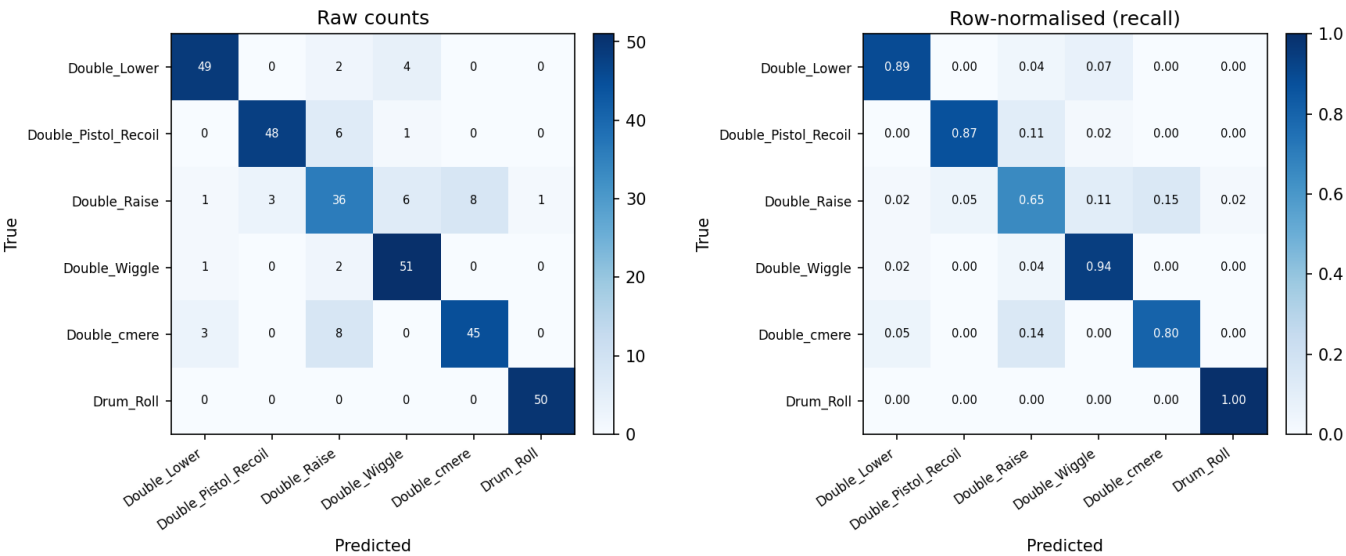
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	795	30	0.9333
2	12_Marcus	14_StephenV2	795	30	0.9000
3	14_StephenV2	17_Raquel	795	30	0.8667
4	17_Raquel	18_Bridgette	795	30	0.7667
5	18_Bridgette	1_Alan	795	30	0.8333
6	1_Alan	2_Alex	795	30	0.9667
7	2_Alex	3_Anghad	792	30	0.8667
8	3_Anghad	4_Daniel	792	31	0.7742
9	4_Daniel	6_Harry	795	30	0.8667
10	6_Harry	8_Jestin	813	30	0.8667
11	8_Jestin	11_Mansh	813	24	0.7917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.907	0.891	0.899	55
Double_Pistol_Recoil	0.941	0.873	0.906	55
Double_Raise	0.667	0.655	0.661	55
Double_Wiggle	0.823	0.944	0.879	54
Double_cmere	0.849	0.804	0.826	56
Drum_Roll	0.980	1.000	0.990	50
macro avg	0.861	0.861	0.860	325
weighted avg	0.859	0.858	0.858	325

Confusion matrix (aggregated across folds)



Variant: BN_GAP_Wide

Conv1D(32,k=8)+BN→Conv1D(64,k=8)+BN→Conv1D(128,k=5)+BN→GAP→Dropout(0.3)→Softmax [LOUO leader]

Trainable params	63,846
LOUO mean ± std	0.8572 ± 0.0851
Folds run	11
Inner-val mode	user
Per-trial norm	zscore

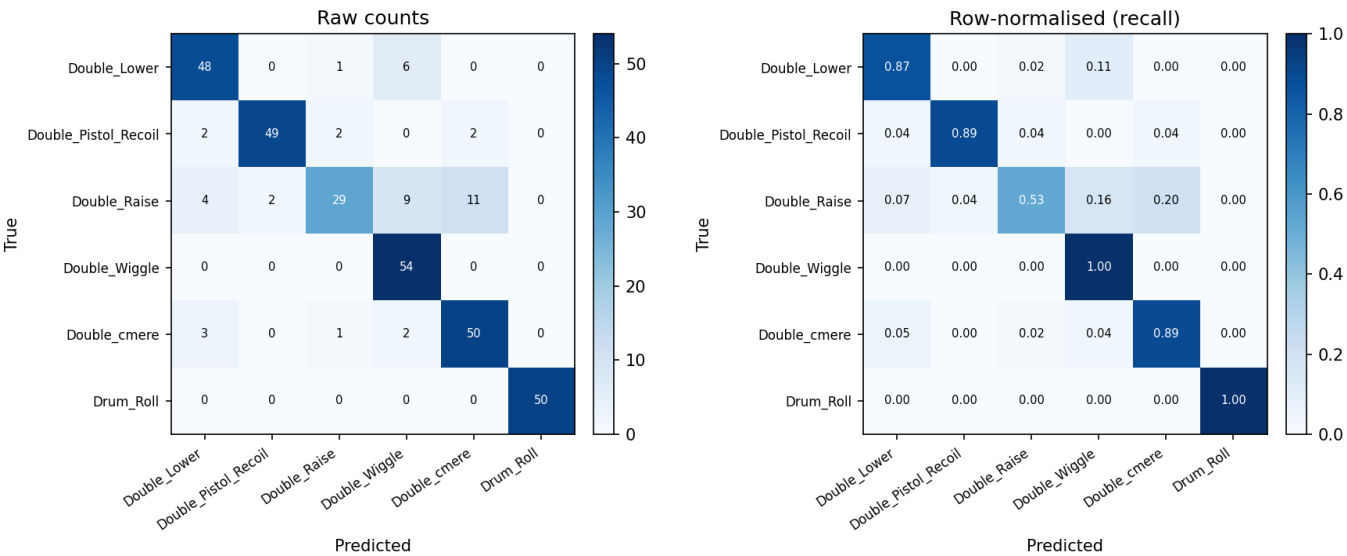
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	795	30	0.8333
2	12_Marcus	14_StephenV2	795	30	0.8667
3	14_StephenV2	17_Raquel	795	30	0.8667
4	17_Raquel	18_Bridgette	795	30	0.8000
5	18_Bridgette	1_Alan	795	30	0.9000
6	1_Alan	2_Alex	795	30	0.9000
7	2_Alex	3_Anghad	792	30	0.9333
8	3_Anghad	4_Daniel	792	31	0.8710
9	4_Daniel	6_Harry	795	30	0.8667
10	6_Harry	8_Jestin	813	30	0.9667
11	8_Jestin	11_Mansh	813	24	0.6250

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.842	0.873	0.857	55
Double_Pistol_Recoil	0.961	0.891	0.925	55
Double_Raise	0.879	0.527	0.659	55
Double_Wiggle	0.761	1.000	0.864	54
Double_cmere	0.794	0.893	0.840	56
Drum_Roll	1.000	1.000	1.000	50
macro avg	0.873	0.864	0.858	325
weighted avg	0.871	0.862	0.855	325

Confusion matrix (aggregated across folds)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	92,966
LOUO mean ± std	0.8441 ± 0.0698
Folds run	11
Inner-val mode	user
Per-trial norm	zscore

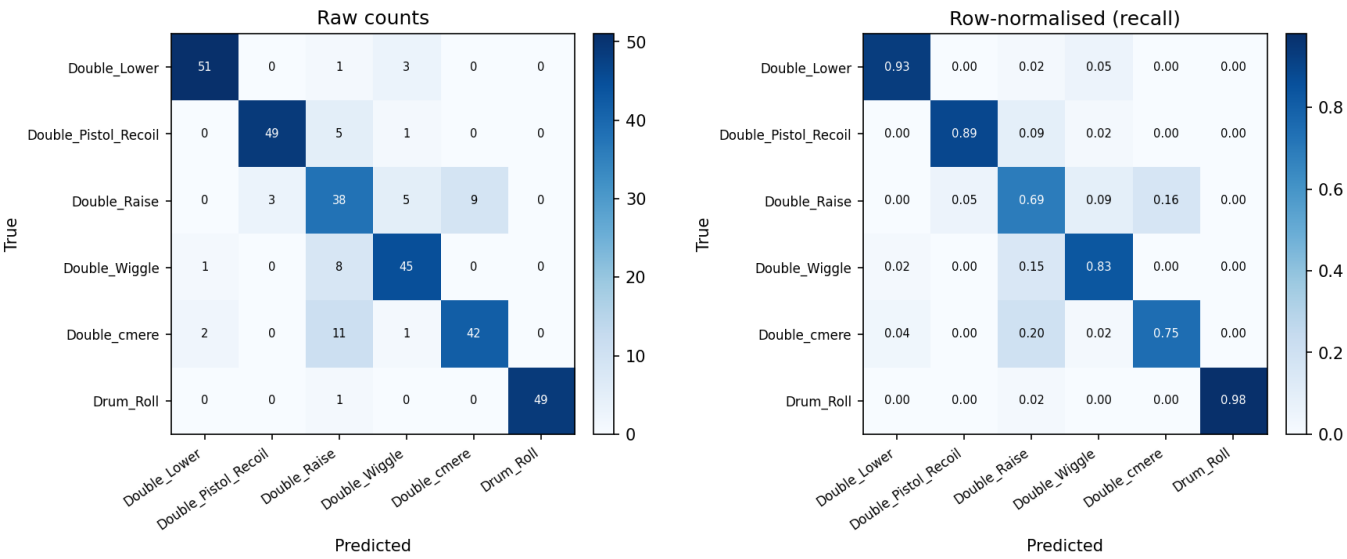
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	795	30	0.9333
2	12_Marcus	14_StephenV2	795	30	0.9000
3	14_StephenV2	17_Raquel	795	30	0.8333
4	17_Raquel	18_Bridgette	795	30	0.7667
5	18_Bridgette	1_Alan	795	30	0.8000
6	1_Alan	2_Alex	795	30	0.9000
7	2_Alex	3_Anghad	792	30	0.7667
8	3_Anghad	4_Daniel	792	31	0.7097
9	4_Daniel	6_Harry	795	30	0.9000
10	6_Harry	8_Jestin	813	30	0.9000
11	8_Jestin	11_Mansh	813	24	0.8750

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.944	0.927	0.936	55
Double_Pistol_Recoil	0.942	0.891	0.916	55
Double_Raise	0.594	0.691	0.639	55
Double_Wiggle	0.818	0.833	0.826	54
Double_cmere	0.824	0.750	0.785	56
Drum_Roll	1.000	0.980	0.990	50
macro avg	0.854	0.845	0.848	325
weighted avg	0.851	0.843	0.846	325

Confusion matrix (aggregated across folds)



Final retrain (deployment model)

Variant	BN_GAP_Wide
Held-out user	— (trained on all users)
Held-out accuracy	—
Training-set size	975 samples (325 before augmentation)
Trainable params	63,846
Final train loss	0.0003
Final train accuracy	1.0000
Model path	— (not saved)
Labels path	— (not saved)
Scaler path	— (per-trial norm in use)